Oral Biology MCQBank

HMHDental

Topic Wise

- Morphology
- Histology

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Tooth development

2- What structure is formed in the crown first?

a- Enamel.
b- Cementum.
c- Pulp.
d- Dentine.

3- Which of the following induces the dental papilla cells to be differentiated into odontoblasts?

a- Stratum intermedium.
b- Reduced enamel epithelium.
c- Inner enamel epithelium.
d- Outer enamel epithelium.

4- No basal lamina is found between the cells of the:

a- Stratum intermedium and inner enamel epithelium.
b- Outer enamel epithelium and the dental sac.
c- Inner layer of Hertwig's sheath and the dental papilla.
d- Inner enamel epithelium and the dental papilla.

5- Which of the following is not a functional activity of the enamel organ:

b- Inducing the differentiation of Odontoblasts.
c- Secretion of enamel matrix.
d- Maturation of enamel.
d- Formation of cementum.

6- Which of the following is not derived from the dental organ:

a- Stellate reticulum
b- Hertwig's epithelial root sheath.
c- Odontoblasts.
d- Ameloblasts.

7-Thestratum intermedium:

a-Inducesdentin formation.
b-Isseparated from the stellate reticulum by a basal lamina.
c-Developsin cap stage.
d-Isimportant for enamel maturation.

8-Thedental organ

a-Develops from cells in the dental follicle.
b-Isacompletely connective tissue structure.
c-Isacompletely epithelial structure.
d-Ishighly vascular, as ameloblasts require an enrichedenvironment.

9-Theenamel cord

a-Isanextension of the dental lamina.
b-Isanextension of the outer dental epithelium.
c-Isanextension of successional dental lamina
d-Extends from the stratum intermedium to the stellatereticulum.

10-Dental follicle cells

a- Differentiate into the ameloblast layer.
b- Differentiate into the odontoblast layer.
c- Are found within the dental organ.
d- Migrate to dentin surface of the root and differentiate into the cementoblast.

11- All of the following is the functions of The Stellate Reticulum except:

a- Keep Space for The enamel development.
b- Act as a reservoir for nutritive materials.
c- Act as a cushioned to protect ameloblasts.
d- Secretes alkaline phosphatase enzyme.

12- The epithelial root sheath of Hertwig disintegrates:

a- After odontoblastic differentiation & before dentin formation.
b- After odontoblastic differentiation & dentin matrix deposition.
c- Before odontoblastic differentiation.
d- After cementum formation.

13- All of the following is a function of dental sac except:

a- Nutrition of enamel organ.
b- Formation of cementum.
c- Formation of periodontal ligament.
d- Formation of supporting alveolar bone

14- The lateral dental lamina gives rise to: a- Deciduous teeth . b- Permanent successors. c- Permanent molars. d- Vestibular lamina. ***15- The stratum intermedium:***

a- Lies between outer enamel epithelium & stellate reticulum.
b- Is a transient structure.
c- Is rich in mucopolysaccharides.
d- Plays important role in enamel calcification.

16- The down growth of an epithelial thickening buccal to the dental lamina is known as:

a- Vestibular lamina.
b- Linguo-alveolar Sulcus.
c- Lateral dental lamina.
d- Successional dental lamina.

17- The dental lamina is induced to proliferate into a tooth bud by the:

a- Basement membrane.
b- Nerve endings.
c- Ecto-mesenchyme.
d- Oral epithelium.

18- Calcified tissues of the tooth are derived from:

a- Ectoderm only.
b- Endoderm only.
c- Mesoderm only.
d- Ectoderm & Mesoderm.

19- A layer of cells that seems to be essential to enamel formation but does not actually secrete the enamel is:

a- Outer dental epithelium.
b- Stratum intermedium.
c- Reduced dental epithelium.
d- Vestibular lamina.

20- The stimulus that initiates the actual formation of enamel matrix seems to be:

a- Reduction of stellate reticulum.
b- Influence of increased vascularization of the dental sac.
c- Disappearance of the cell free zone of dental papilla.
d- Presence of predentin

21- The number of roots that are formed is determined by the:

a- Number of root sheaths developed by the enamel organ.
b- Number of medial ingrowths at the epithelial diaphragm.
c- Number of root sheaths developed by the dental sac.
d- Thickness of the cervical loop.

22- Which of the following is the first process to occur in the sequence of tooth development:

a- Deposition of the first layer of enamel.
b- Deposition of the first layer of dentin.
c- Elongation of the inner dental epithelial cells.
d- Differentiation of odontoblasts.

23-All of the following are involved in the formation of epithelium a tooth except:

a- Epithelial root sheath.
b- Successional lamina.
c- Dental lamina.
d- Vestibular lamina

24- The dental lamina initiating the permanent molars develops:

a-As successional lamina.
b-As a distal extension of the dental lamina.
c-As lateral dental lamina.
d- As vestibular lamina.

25-The cell rests of Malassez are derivatives of:

a- Cervical ameloblasts.
b- Outer enamel epithelium.
c- Dental papilla.
d- Root sheath.

26- Odontogenesis of the primary dentition begins between:

a- The sixth & seventh week.
b- The fifth & sixth week.
c- The seventh& eighth week.
d- The fourth & fifth week.

27- The embryo's stomodeum is lined by:

a- Ectoderm. b- Endoderm. c- Mesoderm. d- Ectomesenchyme.

28- By the 6 week of tooth development, the oral epithelium is:

a- 2-3 layers thickness.
b- 3-4 layers thickness.
c- 4-5 layers thickness.
d- 2-5 layers thickness

30- The ectomesenchnmal cell condensation just beneath the enamel organ is called:

a- Dental sac.
b- Dental follicle.
c- pulp.
d- Dental papilla.

34- Primordium for the permanent dentition appears as an extension of dental lamina into the ectomesenchyme:

a- Lingual to the developing primary tooth germ.
b- Labial to the developing primary tooth germ.
c- Mesial to the developing primary tooth germ.
d- Distal to the developing primary tooth germ.

35- The region where the inner and outer enamel meets at the rim of the enamel organ is known as:

a- Zone of Flection.
b- Cervical loop.
c- Epithelial Diaphragm.
d- Root Sheath of Hertwing's.

36- Which the followings is not a function of stellate reticulum:

a- Protection of the underlying dental tissues.
b- Maintenance of tooth shape.
c- Support the production of enamel.
d- Control the position & number of tooth germs.

37- The epithelial pearls are derived from: a- Dental lamina. b- Dental pulp. c- Dental sac. d- Dental organ.

38- Odontoblasts start their secretory activity: a- Before enamel matrix production. b- After enamel matrix production. c- During enamel matrix production. d- During & after enamel matrix production.

39- Preameloblasts differentiate into ameloblasts: a- After the differentiation of odontoblasts. b- Before the differentiation of odontoblasts.
c- During the differentiation of odontoblasts.
During & after the differentiation of odontoblasts.

40- During the early stages of tooth development three transitory structures may be seen, these structures are:
a- Enamel knot, enamel organ & enamel niche.
b- Enamel knot, enamel cord & enamel niche.
c- Enamel knot, enamel organ & enamel epithelium.
d- Enamel niche, enamel organ & enamel knot.

41- The epithelial rests in the periodontal ligament are derived from: a- Dental pulp. b- Epithelial root sheath of Hertwing's. c- Vestibular lamina. d- Dental sac

42- Appositional stage is confined to:

a- Cap stage. c- Bell stage.

43- Concresense usually occurs with: a- Permanent maxillary molars. b- Permanent maxillary premolars. c- Permanent mandibular premolars. d- Permanent mandibular premolars.

44- Supernumerary roots occur mainly with: a- Permanent third molars. b- Permanent second molars. c- Permanent first molars. d- Permanent second premolars.

45- Enamel pearls occur in: a- Apical third of molars roots. b- Furcation area of molars. c- Cervical third of molars roots. d- Furcation area of premolars.

46-Theprocessofdifferentiationofmesenchymal cellsinto Odontoblastsiscalled:
a-Initiation. b-Induction.
c-Apposition. d-Morphodifferentiation.

47-TheepithelialrootsheathofHertwig'sis composedof:
a-Outerenamelepithelium&innerenamelepithelium.
b-Innerenamelepithelium&stratumintermedium.
c-Outerenamelepithelium&stellatereticulum.
d-Stellatereticulum&stratumintermedium.

48-Innerdentalepithelium:
a-Consistsofasinglelayerofcuboidalepithelialcells.
b-Consistsofasinglelayerofcolumnarepithelialcells.
c-Consistsofasinglelayerofpolyhedralepithelial cells. d-Consistsofasinglelayerofpolygonalepithelial cells.

49-Determinationoftheinitialpositionofthefirst cuspofthetoothduringcrownpatternformation maybeachievedby:
a-Theenamelknot. b-Theenamelcord.
c-Bothenamelknotandcord. d-Noneoftheabove.

50-Beforeenamelformationbegins:
a-Enamelknotdisappearsandenamelcordremains.
b-Enamelknotremainsandenamelcorddisappear.
c-Bothknotandcorddisappear.
d-Bothknotandcordremain.

51-Concerningthestellatereticulumoneofthe followingstatementsiswrong:
a-Occupiesthecentralportionofthedentalorgan.
b-Liesbetweenouterandinnerdentalepithelia.
c-Synthesizesandsecretesglucoseaminoglycans.
d-Becomespolyhedralinshape.

52-Thedentalsacis:
a-Thecondensationoftheectomesenchymalcells surroundingthedentalorgan.
b-Thecondensationoftheectomesenchymalcells surroundingthedental papilla.
c-Thecondensationoftheectomesenchymalcells surroundingthedentalorgan anddentalpapilla.
d-Noneoftheabove

53- The main difference between bell and apposition stage is:

a- The formation of first layer of E.
b- The formation of first layer of D.
c- The formation of first layer of C.
d- The formation of first layer alveolar bone.

54- Concerning the stratum intermedium, one of the following statements is incorrect:

a- It is composed of 2-3 layers of squamous epithelial cells.
b- It is differentiated between the inner enamel epithelium and stellate reticulum.
c- It is connected to each other and to the cells of stratum intermedium by desmosomes.
d- It is essential for enamel formation as it contains acid phosphatase enzyme.

55- Which one of the following functions of the inner dental epithelium is wrong?

a- It exerts an organizing influence on the undifferentiated cells of the dental papilla to differentiate into ameloblasts.
b- It shares in the transport of the nutritive materials.
c- It determines the future incisal or occlusal crown pattern.
d- It lays down enamel matrix and helps in its mineralization

56- The epithelial root sheath of Hertwig can't be seen as a continuous layer in the developing root because:

a- The rapid epithelial sheath proliferation.
b- The dentin formation.
c- The rapid epithelial sheath destruction after dentin formation.
d- All of the above.

57- Epithelial rests of Malassez are:

a- Epithelial strands and may undergo degeneration only.
b- Epithelial remnants of the epithelial root sheath of Hertwig only.
c- Network or isolated islands in the P.L. only,
e- All of the above.

58- If Hertwig's epithelial root sheath does not disintegrate:

a- No cementum will be formed on the radicular dentin.
b- It is likely that some of the enamel will be resorbed.
c- The apical end of the junctional epithelium will lie occlusal to the cemento-enamel junction.
d- Predentine will form on the outer surface of the radicular dentin.

59-The reduced enamel epithelium is derived from:

a- Maturative ameloblasts and stratum intermedium.
b- Stratum intermedium, stellate reticulum and outer dental epithelium.
c- Stratum intermedium, stellate reticulum and inner epithelium.
d- Protective ameloblasts, stratum intermedium, stellate reticulum and outer dental epithelium.

60-Which of the following is not component of the enamel organ?

a- Ameloblasts.
b- Odontoblasts.
c- Outer enamel epithelium.
a- Stratum intermedium.

61- Which of the following indicates the major components of a tooth germ?

a- The dental lamina, the stellate reticulum and the stratum intermedium.
b- The dental organ, the dental papilla and the dental sac.
c- The dental pulp, the dental sac and the dental lamina.
d- The cervical loop, Hertwig's sheath and the inner enamel epithelium.

62- Which of the following is not ectodermal in origin?

a- Epithelial rests of Malassez.
b- Anterior pituitary gland.
c- Stellate reticulum.
d- Enamel spindles.

63- Which of the following appear FIRST in a developing tooth?

a- Predentin.
b- Cementoid.
c- Tome's process.
d- Secondary dentin.

64- The last hard dental tissue to be deposited is:

a- Enamel.
b- Outer layers of cementum.
c- Mantle dentin.
d- Primary dentin.

65- Which of the following is associated with an erupted tooth?

a- Outer Enamel epithelium.
b- Cervical loop.
c- Epithelial rests of Malassez.
d-Stratum intermedium..
d)Dental organ, dental follicle, dental lamina.

66- Which of the following is not true of Hertwig's epithelial root sheath?

a- It lacks a stellate reticulum.
b- It is derived from stomodeal ectoderm.
c- Its remnants form epithelial rests.
d- It remains intact until a layer of cementum has been formed.

67- Which of the following contain blood vessels?

a- Cartilage.
b- Enamel.
c- Dental lamina.
d- Dental sac.

68- Which group of structures is entirely derived from the dental sac?

a- Dentin, pulp, cementum.
b- Cementum, alveolar bone, pulp.
c- Cementum, enamel, pulp.
d- Periodontal ligament, cementum, alveolar bone.

69-The development of the root begins when

a) The inner and outer dental epithelium reaching the future C.E.J.
b) The crown is completely formed.
c) The development of the supporting bone begins.
d) The development of the periodontal ligament begins

70- The function of the inner dental epithelium is……..

a) Exerts organization influence on the dental papilla cells.
b) Lays down dentin matrix
c) Protects the enamel after eruption.
d) Provides the enamel organs with alkaline phosphatase

71-The tooth germ is composed of

a) Dental papilla, dental follicle, dental organ.
b) Dental follicle, dental lamina, vestibular lamina.
c) Dental papilla, dental lamina, dental organ

72-The apposition stage of tooth development begins

a) When the root of the tooth begins to develop.
b) When the first layer of dentine is deposited.
c) When odontoblasts start to differentiate

known as: a- Dental lamina. b- Dental papilla. c- Enamel organ. d- Dental follicle. ***81- Which of the following is not part of the tooth germ?*** a- Dental papilla. b- Dental follicle. c- Dental lamina. ***73- What statement about epithelial diaphragm is Not*** d- Enamel organ.

true?

a- It produces narrowing of the wide cervical opening.
b- Its proliferation is accompanied by proliferation of pa- cells adjacent to diaphragm.
c- It's growing end is located coronal to the root sheath.
d- It grows in a horizontal plane.

74- The cells of the stratum intermedium:

a- Have a stellate outline.
b- Are not yet differentiated in the bell stage.
c- Are in direct contact with the outer dental epithelium.
d- Are characterized by high alkaline phosphatase enzyme activity.

75-The dental follicle:

a- Is an epithelial structure.
b- Give rise to the radicular pulp.
c- Is not present in permanent teeth.
d- give rise to cementum.

76-If the process of morphogenesis is disturbed:

a- Changes will occur in the shape of crown and root.
b- Enamel formation will be abnormal.
c- Dentin formation will be abnormal.
d- Absence of cementum.

77- The mesenchymal cells lying adjacent to the inner enamel epithelium:

a- Differentiate into ameloblasts.
b- Differentiate into odontoblasts.
c- Are under the influence of stellate reticulum.
d- Become markedly folded.

78- The dental lamina:

a- Breaks at the cap stage.
b- Has no branching.
d- Has no remnants.

79- Advanced bell stage of development refers to:

a- Histodifferentiation , morphodifferentiation. b- Initiation & Proliferation. c- Mesenchymal condensation. d- Apposition of dental tissues.

82- The dental sac plays an important role in the formation of all of the following except:

a- Periodontal ligament.
b- Dentin of the root.
c- Alveolar bone proper.
d- Cementum.

83- After the crown of the tooth is formed, the dental papilla is called:

a- Dental pulp. b- Pulp stone. c- Dental sac. d- Dentin.

84- In a developing tooth, if Hertwig's sheath and the epithelial diaphragm were failed to form, the resulting tooth would exhibit:

a- No radicular dentin.
b- Malformed or fused roots.
c- Acellular, but no cellular, cementum.
d- Defective enamel in the cervical region.

85- The reduced enamel epithelium:

a- Lacks cells derived from the stellate reticulum.
b- Contains protective ameloblasts.
c- Overlies immature enamel which has not yet reached its full thickness.
d- Does not cover enamel located at the cervical end of the crown.

86- Which of the following is a component of tooth germ?

a- Lateral dental lamina.
b- Successional lamina.
c- Vestibular lamina.
d- Dental sac.

87-The origin of enamel is:

a- Endoderm.
b- Mesoderm.
c- Ectoderm.
d- Ectomesenchyme.

88-Which statement about early dental development is incorrect?

Cap stage is accomplished by unequal growth in different parts.
In the cap stage the inner and outer enamel epithelial cells are completely alike.
The transition from cap stage to bell stage is particularly marked by the full differentiation of stellate reticulum.
The dental papilla and the dental sac develop in the cap stage.

89- The dental sac gives rise to:

a- Cementoblasts and periodontal ligament.
b- Periodontal ligament and dental pulp.
c- Dental pulp and attachment epithelium.
d- Odontoblasts and Cementoblasts.

Enamel

1- Ameloblasts having ruffled borders are:

f- Secretory ameloblasts.
g- Maturative ameloblasts.
h- Protective ameloblasts.
i- Transitional ameloblasts.

2-Which of the following is not a feature of enamel?

a- Contour lines of Owen.
b- Cross striations.
c- Prism sheath.
j- Perikymata.

3-Maturative ameloblasts are characterized by having:

a- A distally positioned nucleus.
b- A highly basophilic cytoplasm.
c- A highly-developed Golgi complex.
d-A distal cell membrane with many infoldings.

4- Which of the following does not occur during maturation of enamel?

a- Influx of calcium and phosphorus.
b- Growth of hydroxyapatite crystals.
c- Increase in organic content.
d-Decrease in the water content.

5- Incremental lines are a result of variation in:

a- Proliferation.
b- Histodifferentiation.
c- Morphodifferentiation.
d- Rhythmic apposition.

6- The prismatic structure of enamel is due to:

riation in the density of organic material.
riation in crystal orientation.
hythmic pattern of apposition.
d- The periodic change in prism direction.

7- The prismless layer of enamel:

ound only in primary teeth.
es not contain prism & prism sheaths.
ound near the amelo-dentinal junction.
ntains no hydroxyapatite.

8- Enamel tufts:

a- Contain cellular processes.
b- Can pass to dentin.
c- Are seen in decalcified sections.
d- Are grass like structure

9- During the maturation stage of enamel formation, which of the following does not occur in the ameloblastic layer?

a- Water absorption.
b- Protein absorption.
c- Change in cellular organelles.
d- Secretion of enamel matrix.

10- When prismless enamel is formed, the secretory ameloblast does not contain:

a- Rough endoplasmic reticulum.
b- Tome's process.
c- Nucleus.
d- Secretory granules.

11- Which of the following is true:

d- Enamel is the weakly mineralized tissue.
e- Enamel is the highly mineralized tissue.
f- Enamel is the highly permeable tissue.
d-Enamel is a non brittle tissue.

12-Neonatal lines are present in:

a-All permanent teeth.
b-Permanent canines.
c-All deciduous & first permanent molar.
d- Premolars.

13- Organic components of enamel are primarily removed from enamel by:

a- Secretory ameloblasts.
b- Maturative ameloblasts.
c- Transitional ameloblasts.
d- Enamel spindle.

14- The main difference between the rod and the interrod region is:

a- Crystal size.
b- Crystal orientation.
c- Crystal composition.
d- In the nature of the organic component.

15- The striae of Retzius:

a- Are parallel to the enamel rods.
b- Are parallel to the surface of the enamel in the cervical region.
c- Are not seen in the cuspal one third of the enamel.
d- Would be seen as concentric rings in a horizontal section of the crown.

16-Enamel is composed of:

a) Rod, interrod & rod sheath.
b) Only enamel prisms.
c) Needle shaped crystals.
d) Prism sheath.

17-The extensions from secretory ameloblast cells to enamel rods are called:

a) Enamel tuft.
b) Tomes processes.
c) Enamel spindles.
d) Enamel lamellae.

18- Striae of Retzius:

a) Result from enamel crack.

b) Separate prenatal from postnatal enamel.

c) Are due to rhythmic apposition of enamel.

d) Are hypermineralized structure.

19- Enamel Rods:

a) Their number varies in different teeth.

b) Have an average width of 1um.

c) Are Present at the enamel surface.

d) Have a straight course at the cusps of teeth.

20- Enamel is formed of:

a- 76% minerals & 24% organic materials and water.

b- 96% minerals & 4% organic materials and water.

c- 86% minerals & 14% organic materials and water.

d- 99% minerals & 1% organic materials and water.

21- which of the following is absent in all premolars:

a- Enamel spindle.

b- Enamel tuft.

c- Neonatal line.

d- Enamel lamella.

22- Cross striations in enamel:

a- Are an optical phenomena.

b- Result from abrupt change in environment before and after birth.

c- Result from mineralization disorder.

d- Appear to be periodic bands at 4 microns interval across the rod.

23- The prism sheath:

a- Is a dense continuous sheath separating rod from inter-rod.

b- Result from abrupt change in crystals orientation.

c- Is more mineralized than rod and inter-rod.

d- Found in prismless enamel.

24- The complex arrangement of rods over the cusps of teeth is termed: a- Prismless enamel. b- Gnarled enamel. c- Prenatal enamel.

d- Postnatal enamel.

25- The origin of enamel is:

a- neural crest cells

b- Ectoderm.

c- Mesoderm.

d- Ectomesenchyme.

26- Hypocalcified areas in enamel are:

a- Enamel spindles.

b- Dentino-enamel junction.

c- Interprismatic substances.

d- Enamel tufts.

27- Prismless enamel:

a- Is less mineralized than prismatic enamel.

b- Is found only in the cervical area of permanent teeth.

c- Is harder than prismatic enamel.

d- Found near the dentino-enamel junction.

28- The perikymata:

a- Is an optical phenomena resulting from changes in direction of enamel rods.

b- Is a highly mineralized outer enamel surface.

c- Is formed of transverse grooves, represents an outer manifestation of

brown stria of Retzius.

d- Found at the cusp tip. ***29- What is the result of cracks before tooth eruption:*** a- Enamel lamella type A. b- Enamel lamella type B.

c- Enamel lamella type C.

d- Enamel tufts.

30- The primary enamel cuticle is formed by the:

a- Outer dental epithelium.

b- Ameloblasts just before tooth eruption.

c- Fusion of oral epithelium and reduced enamel epithelium.

d- Ameloblasts after the enamel has been completed.

31- The surface enamel:

a- Is similar to the subsurface enamel.

b- Is softer than subsurface enamel.

c- Contains less mineral than subsurface enamel.

d- Highly mineralized than subssurface enamel.

32- The prism sheath:

a- Is a continuous sheath.

b- Is more calcified than the enamel rod.

c- Is less calcified than the enamel rod.

d- Is equally calcified when compared to the enamel rod.

33- The predominant inorganic portion of enamel is: a- Calcium. b- Carbonate. c- Phosphate.

a- Fluoride.

34- Incremental lines of Retzius form grooves on the surface of the enamel termed:

a- Perikymata.

b- Neonatal lines.

c- Gnarled enamel.

d- Enamel cuticle

35- Which of the following is poorly calcified enamel?

a- Gnarled enamel.

b- True enamel lamellae.

c- Perikymata and gnarled enamel.

d- Interprismatic substance

36- The cross striations characteristic of enamel rods represent:

a- Incremental growth pattern.
b- The junctional complex of ameloblasts.
c- Hunter-Schreger bands.
d- The inter rod region.

37- Enamel spindles:

a- Are areas of prismatic enamel.
b- Are present at the occlusal surfaces of enamel.
c- Are the terminal ends of odontoblastic processes.
d- Are located at the cemento-enamel junction.

38- Which of the following is a true statement about in: the organic matrix of the newly synthesized enamel?

a- It represents 30% by weight of the organic content.
b- It has a high content of amelogenins.
c- It contains collagen fibers.
d- It has a high inorganic content.

39- Which of the following is the source of nutrient supply to the secretory ameloblasts during early crown formation?

a- Subodontoblastic plexus of capillaries.
b- Central blood vessels in the dental papilla.
c- Dental sac capillaries adjacent to the epithelial diaphragm.
d- Dental sac capillaries adjacent to the outer dental epithelium.

40- Tome's processes:

a- Contain mitochondria, but lack Golgi saccules and rough endoplasmic reticulum.
b- Lack secretory granules.
c- Indent the immature enamel to produce a serrated outline on its appositional surface.
d- Are located at the proximal end of the ameloblast.

41- Enamel spindles are formed by:

a- Odontoblastic process.
b- Ameloblastic process.
c- Hypocalcified rods.
d- Cracks.

42- Neonatal lines are found in all the following except:

a- Enamel of primary incisors.
b- Enamel of permanent first molar.
c- Enamel of permanent second molars.
d- Enamel of primary molars.

43- Hypocalcified ectodermal structures in enamel are:

a- Spindles and lamellae.
b- Odontoblastic processes and tufts.
c- Tufts and spindles.
d- Lamellae type A,and tufts.

44- Which of the following most correctly describes the significance of Hunter Schreger bands? It indicates variation in size of enamel rods. It reflects variations in the length of enamel rods. It represents enamel. It is an optical effect that is not related to variations in enamel structure.

45- The orientation of hydroxyapatite crystals in enamel, the greatest crystal density would be present

a- enamel lamellae.
b- The interrod region.
c- The prismless layer.
d- Gnarled enamel.

46- Perikymata:

a- Are oriented parallel to the long axis of the crown (tooth).
b- Are surface manifestation of Retzius lines.
c- Are only located in prismless enamel.
d- Produce the scalloping seen along the DEJ.

47- Unmineralized structures seen in the enamel are :

a- Cross striations.
b- Enamel tufts.
c- Incremental lines of Retzius.
d- Enamel spindles.

48- All of the following statements about the incremental lines of Retzius in enamel are true except:

a- They run as concentric lines over the cusp.
b- Perikymata are outer manifestations of it.
c- They are formed as result of rhythmic apposition.
d- They do not reach the cervical region.

49-Incremental lines of Retzius appear:

a- In dentin.
b- In enamel.
c- In cementum.
d- In bone

50-Primary enamel cuticle is:

a- Afibrillar cementum.
b- Precipitate of salivary proteins.
c- Relatively structureless layer of enamel.
e- A last product of ameloblasts.

51- Hunter-Schreger bands have all the following characteristics except:

a- They are best seen in reflected light.
b- They are alternating dark and light bands
c- They are most likely an optical phenomenon
d- They are found at the cusp tip.

52- Enamel tufts and enamel lamellae resemble each other in that both:

a- Are formed prior to tooth eruption only.
b- Extend over full thickness of the enamel.
c- Are hypermineralized structures.
d- Are hypomineralized structures.

53-Enamel is by weight:

a- 70% mineral, 26% organic matrix, 4% water.
b- 76% mineral, 20% organic matrix, 4% water.
c- 25% mineral, 12% organic matrix, 3% water.
d- 95% mineral, 1% organic matrix, 4% water.

54- All of the following structures in enamel are ectodermal origin except:

a- Enamel spindle.
b- Enamel prisms.
c- Enamel lamellae (type A).
d- Enamel tufts.

55- Enamel is:

a. An ectodermal tissue covering the anatomical crown.
b. A mesodermal tissue covering the crown.
c. An endodermal tissue.
d. A mesenchymal tissue.

56- Gnarled enamel has the following criteria except:

a. Found under cusp tips and incisal ridges.
b. The course of the enamel rods is more straighter.
c. The enamel rods become twisted and braided.
d. Found to give maximum strength to the areas subjected more to the masticatory forces.

57- The peripheral part of the E rod is called:

a. Interrod region. b. Rod sheath.
c. Rod region. d. Key hole.

58- Concerning the rod sheath, one of the following statement is incorrect:

a. It occupies the peripheral part of the E. rod.
b. It forms the border where crystals of the rod and interrod enamel meet.
c. It is highly calcified than the rod.
d. It has a higher organic content thus resists decalcification more than the rod.

59- The interrod enamel is:

a. More calcified than the E. rod.
b. Less calcified than the E. rod.
c. The same calcification as the E. rod.
d. Organic in nature.

60-Crossstriationsorshortincrements:

a.Representtherhythmicappositionoftheamelo blasts.
b.Representalternatingconstrictionandexpansionof rodsatandin-betweenthestriations.
c.Representthedailyrateofsecretoryactivityofthe ameloblast.
d.Alloftheabove.

61-Theformationoftheenamelmatrixbetweentwo shortincrementsneeds:

a.Oneday. b.Twodays.
c.Threedrays. d-Fourdays.

62-Neonatallineisallthefollowingexcept:

a.Isoneofthebrownstriaeof accentuated.
b.Ispresentindeciduousteethandallpermanent molars.
c.Separatesenamelformedbeforebirthfromthat formedafterbirth.
d. It is due to abrupt change in nutrition and environmentalconditionatbirth.

63-ThequalityoftheprenatalEnamel:

a.Isbetterthanthatofthepostnatal .
b.Isnotasbetterasthatofthepostnatal .
c.Isequaltothequalityofpostnatal .
d.Notrelatedtothequalityofpostnatal .

64-Thedentinoenameljunction(DEJ):

a.Appearstobescallopedwithconvexitiesofthe scallopsdirectedto .
b.Appearstobescallopedwithconvexitiesofthe scallopsdirectedto .
c.Appearstobescallopedwithconvexitiesofthe scallopsdirectedto cementum.
d.Noneoftheabove.

65-ThefirmattachmentbetweenEandDisdueto:

a.UndercutscreatedbythescallopsoftheDEJ.
b.InterdigitationattheDEJbetweenfibrilsof firstformedlayerofDandthatof firstformedlayerof .
c.Interdigitationbetweenthehydroxyapatite crystalsthataredepositedduring mineralizationofEandD.
d.Alloftheabove.

66- E. lamellae are all the following except :

a. Thin leaf or sheet like structures.
b. Extend from the E surface and may extend to D.
c. Demonstrated in the longitudinal ground section only.
d. Hypo or unmineralized structures

67- Enamel spindles:

a. They are odontoblastic cell bodies pass across the DEJ into the E.

b. They are extended between the preameloblasts after crown enamel is formed

c. They are most common beneath cusps.

d. They are best demonstrated in longitudinal decalcified section.

68- Rodless Enamel:

a. Occurs in the outermost 70 microns of the E.

b. Their crystals are aligned at right angel to the surface

c. Found most common at the cervical areas of E surface.

d. All of the above.

69- Perikymata are:

a. Vertically oriented wave like grooves.

b. The external manifestation of the brown striae of Retzius.

c. Numerous at the incisal or cusp regions.

d. Parallel to the AEJ

70- Which of the following tissues begins formation first:

a. Enamel. b. Dentin.

c. Pulp. d. Cementum.

71- Attrition is characterized by:

a. Being a condition of normal wear of the occlusal and proximal surfaces.

b. A chemical loss of occlusal tooth structure, but not from dental caries.

c. A loss of crown structure on the occlusal surface by the chemical action of acids.

d. Loss of tooth structure by abrasion.

72- In mature dental tissues, the only tissue whose cells are lost is:

a. Dentin. b. Enamel.

c. Cementum. d. Pulp.

73- Incremental lines of Retzius appear:

a. In dentin and follow the appositional pattern.

b. In enamel and follow the appositional pattern.

c. In cementum and follow the appositional pattern.

d. As lines in E running at right angles to the E surface

74- Enamel is by weight:

a. 88-90 % mineral, 10-12 % organic matrix and water.

b. 76% mineral, 20% organic matrix, 4 water.

c. 25% mineral, 12% organic matrix, 3% water.

d. 96% mineral, 1-2% organic material, 2-3% water.

75- Which of the following is not true about Enamel rod ?

a. They run from the dentinoenamel junction to the surface.

b. Length of all enamel rods is equal.

c. They are the basic structural units of E.

d. They may be hexagonal ,rounded, oval or keyhole shape in cross section .

76- Which of the following statements about E . is false?

a. The heads of enamel rods are directed cervically, the tails occlusally.

b. Rod sheaths are relatively hypocalcified areas surrounding the rods.

c. Human E rods in cross section have an outline similar to a keyhole.

d. Hydroxyapatite crystals run parallel to the long axis of the rods and perpendicular to it in the interrod region .

77- It is particularly difficult to finish the wall of a cavity preparation in cuspal enamel because of the presence in that area of:

a. Tufts. b. Horizontal E. rods.

c. Gnarled E. d. Spindles.

78- Enamel tufts and Enamel lamella resemble each other, in that both:

a. Are formed after to tooth eruption.

b. Extend over full thickness of the enamel.

c. Are hypermineralized structures.

d. Are oriented in a vertical direction perpendicular on DEJ.

79- All of the following statements about the Neonatal lines of Retzius in Enamel are true except:

a. In longitudinal section they run obliquely, relative to D.E.J.

b. Perikymata in newly erupted teeth are manifestations of the lines of Retzius.

c. They are formed as a result of rhythmic variations in the calcification.

d. Under no conditions can a line of Retzius to be a neonatal line.

80- E. has no possibility of self repair because:

a. It has a small percentage of organic content.

b. Its formative cells are lost once it is fully formed and the tooth erupted.

c. It is essentially a keratin tissue and has no blood vessels.

d. It has no direct connection with the active cells of the pulp tissue.

81- At the time Enamel matrix is first formed in a tooth:

a. No predentin has yet been formed.
b. Root of the tooth is largely covered with acellular cementum.
c. Nuclei of ameloblasts move to non secreting end of the cell.
d. Cells of stellate reticulum near the cervical portion of the tooth have lost their nuclei.

82- Hypomineralized structures which extend from the D.E.J to the surface of the Enamel are called:

a. Hunter schreger bands.
b. E. spindle.
c. Incremental lines of Retzius.
d. Enamel lamellae.

83- Maturation of Enamel is characterized by a percentage increase in inorganic content and a percentage:

a. Increase in water.
b. Decrease in water and organic content.
c. Increase in water and decrease in organic content.
d. Decrease in water and increase in organic content.

84- The stimulus that initiates the actual formation of Enamel matrix seems to be the:

a. Presence of predentin.
b. Reduction of stellate reticulum.
c. Influence of increased vascularization of the dental sac.
d. Reversal of functional polarity of ameloblast.

85- In a routine demineralized histological section of an adult tooth, the area representing Enamel is generally optically clear because:

a. The mounting medium has the same refractive index as enamel.
b. E is very brittle and thus breaks off during sectioning.
c. Ordinary stains don't stain the organic matrix of E.
d. Enamel is highly calcified.

86- A layer of cells that seems to be essential to enamel formation but does not actually secrete the enamel is:

a. Inner dental epithelium.
b. Stratum intermedium.
c. Reduced dental epithelium.
d. Vestibular lamina.

87- Which of the following are poorly calcified Enamel:

a. Gnarled E .and E rods.
b. E. lamellae and incremental lines.
c. Perikymata and gnarled E.
d. E. rods and incremental lines.

88- Enamel matrix is laid down by ameloblasts:

a. Finally contains about 80% by weight inorganic matter.
b. Includes organic matter that is primarily collagen.
c. Persists as a vital tissue through the life of a tooth.
d. None of the above.

89- Enamel tufts:

a. Begin on the outer edge of E .
b. Are developmental defects and are worn away by masticatory forces.
c. Have a brush like appearance and are hypo calcified areas in E.
d. Represent hypercalcified E. rods.

90- The rod sheath:

a. Has cross striations.
b. Is more calcified than the E. rod.
c. Is less calcified than the E. rod.
d. Has lesser amount of organic matter than the E. rod.

91- Ionic exchanges between erupted Enamel .and Saliva:

a. Don't occur as the formative cells of the E .are lost once it is formed.
b. Occur only for two to five years following eruption.
c. Cease at puberty.
d. Occur throughout life.

92- The predominant organic portion of mature Enamel. is:

a. Calcium.
b. Fluoride.
c.Enamelin.
d. Phosphate.

93- Fluorine deposition in Enamel:

a. Increase with age.
b. Decrease with age.
c. Sometimes increases or decreases with age.
d. Not affected by age.

94-Which of the following organic components are present in Enamel:

a. Carbohydrates.
b. Fat.
c. Proteins.
d. All of the above.

95- The rodless layer of enamel:

a- Is found only in primary teeth.
b- Does not contain rod and rod sheaths.
c- Is found near the amelo-dentinal junction.
d-Contains no hydroxyapatite.

96- Hunter-Schreger bands are:
a-found in the outer surface of enamel.
b-run parallel to the striae of Retzius.
c-best visualized in transverse section of the crown.
d-reflect periodic changes in prism direction of enamel.

97- The direction of enamel rods in general is:
a-parallel to the outer surface of enamel.
b-parallel to dentine.
c-perpindicular to the amelo-dentinal junction.
d-random.

98- Which of the following contains the highest concentration of hydroxyapatite:
a-cementum.
b-enamel.
c-primary dentine.
d-secondary dentine.

99-Enamel prism of the cusp tip passes:
a-inwavy course.
b-ins-shape course.
c-instraight course.
d-intwisted course.

100-Developmental enamel lamellae is:
a-type A enamel lamellae.
b-type B enamel lamellae.
c-type C enamel lamellae.
d-accidental enamel crack.

101- The structures which are not seen in a decalcified section are:
a- Dentinal tubules.
b- Incremental lines of Retzius.
c- Secondary dentin.
d- Cellular cementum.

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Dentin

1-Interglobular spaces:

a- Do not contain dentinal tubules.

b- Do not contain type I collagen.

c- Are formed during the course of globuc- mineralization.

d- Are seen only in secondary dentin.

2- Which would not be present in a ground section?

a- Inorganic crystals.

b- Odontoblast.

c- Interglobular space.

d- Dentinal tubules.

3- A major difference between secondary and primachanges? dentine is that:

a- There is no predentine.

b- There is no hydroxyapatite.

c- There is no collagen fibers.

d- The dentinal tubules pass in wavier course throu***12-*** secondary dentine.

4- The interglobular dentin is found:.

a- Near the amelo-cemental junction.

b- In the root dentin.

c- Near the amelo-dentinal junction.

d- Near the dentino-cemental junction.

5- Which of the following is MOST likely to be found d- ***the dentinal tubules:***

a- Tomes' fibers.

b- Oxytalan fibers.

c- Tomes' process.

d- Thick collagen fibers.

6- Primary dentin differs from both secondary aof the dentin. ***reparative dentin in that:***

a- Its matrix is calcified.

b- Its inorganic component is hydroxyapatite.

c- It usually has a greater number of tubules per ua- area.

d- It is produced by mesenchymal cells.

7- Which of the following is not present within dentin?

a- Cell bodies.

b- Hydroxyapatite.

c- Collagen type I.

d- Incremental lines

8- Which of the following is true of peritubular dentin ? ***:***

a- Is less mineralized than intertubular dentin.

b- May be considered as intratubular dentin.

c- Is best visualized in decalcified sections.

d- Is found in the interglobular dentin.

9- Which of the following is not present in odontoblasts?

a- Golgi saccules and vesicles.

b- Rough endoplasmic reticulum.

c- Desmosome-like junctions.

d- Numerous processes forming a striated border.

10- Human primary dentin contains:

a- Mantle & circumpulpal dentin.

b- Tomes' process.

c- Cell bodies.

d- Atubular dentin.

11- Which of the following is considered as age changes?

a- Transparent dentin.

b- Cellular cementum.

c- Primary dentin.

d- Enamel spindle.

12- Which of the following is true: ___

a- A layer of predentin is found between primary and secondary dentin.

b- Dentinal tubules do not pass through interglobular dentin. dentin contain

c- The physiological secondary dentinal tubules.

Many large collagenous fiber bundles are found in the dentinal tubules.

13- Reparative dentin:

a- Contain regular and numerous dentinal tubules.

b- Obliterates the apical foramen.

c- Forms on localized regions of the pulpal surface of the dentin.

d- Formation is induced by the cells of the inner dental epithelium.

16- Collagenous fibers are not found in:

a- Mantle dentin.

b- Reparative dentin.

c- The outer prismless layer.

d- The dental sac.

17- Obliteration of the dentinal tubules by mineral deposits:

a- Occurs only when the dentin becomes exposed to the oral fluids.

b- Produces dead tracts.

c- First occurs in predentin.

d- Is seen in sclerotic and transparent dentine.

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18- Tome's granular layer is:

a- An extension of odontoblast processes into enamel.
b- A region lacking dentinal tubules.
c- Following the incremental lines.
d- Derived from the epithelial rests of Malassez.

20- Which of the following is wrong?

a- The formation of primary dentin continues throughout life.
b- Primary dentin contains a significantly higher number of tubules per unit volume.
c- Reparative dentin forms as a response to injury.
d- Dead tracts are more common in secondary than in primary dentin.

21- Which of the following has the highest mineral content?

a- Predentin.
b- Peritubular dentin.
c- Intertubular dentin.
d- Cementum

23-Cementum is a tissue that:

a- provides attachment of periodontal ligament fibers.
b- is easily resorbed than bone.
c- it contains blood vessels.
d- it contains neural receptors.

24-Cementum deposition cervically on enamel surface is due to:

a- premature degeneration of epithelial root sheath of Hertwig.
b- loss of reduced enamel epithelium.
c- persistence of epithelial root sheath of Hertwig.
d- cementoblasts coming from dental sac.

25-Neonal line is present in:

a- enamel and cementum.
b- dentin and cementum.
c- enamel and dentin.
d- none of the above.

26-Obliteration of dentinal tubules by minerals will result in:

a- formation of dead tracts.
b- formation of secondary dentin.
c- formation of sclerosed dentin.
d- formation of irregular or reparative dentin.

27-Age changes will increase:

a- dentin sclerosis.
b- interglobular dentin.
d- in cementum thickness.
d- Tome's granular layer

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Cementum

1- Cementum may cover part of the enamel due to:

a- Premature differentiation of cementoblasts.
b- Destruction of the reduced enamel epithelium covering this part.
c- Malfunctioning of the maturative ameloblasts.
d- Premature destruction of the epithelial root sheath of Hertwig.

2- Which of the following cell is responsible for cementogenesis?

a- Ameloblast.
b- Osteoblast.
c- Cementoblast.
d-Odontoblast.

3- Which of the following is true about cementum? It:

a- Provides nutrients to secondary dentin in the root.
b- Provides for the attachment of principal periodontal fibers.
c- Contains neural receptors.
d- Is more easily resorbed than bone.

4- Which of the following is not a feature of cellular cementum?

a- Lacunae and canaliculi.
b- Continuous remodeling.
c- Sharpey's fibers.
d- Incremental lines.

5- Cementum may be deposited on the surface of enamel as a consequence of:

a- Early degeneration of Hertwig's sheath.
b- Injury to the reduced enamel epithelium.
c- Late degeneration of Hertwig's sheath.
d-During passive eruption.

6- Cementoblasts:

a- Are derived from the dental papilla.
b- Secrete both collagen and glycosaminoglycans.
c- Are multinucleated cell.
d- May becomeincorporatedintoacellular cementum.

7- Which of the following is not present inside both cellular and acellular cementum:

a- Cementoblasts
b- Hydroxyapatite crystals.
c- Incremental lines.
d- Sharpey's fibers.

8- Which of the following does not increase normally with age?

a- Pulp stones.
b- Dentin sclerosis.
c- Interglobular dentin.
d- Cementum thickness.

9-human teeth contain varities of cementum:

a- acelllular extrinsic fiber cementum.
b- cellular intrinsic fiber cementun.
c- mixed stratified cementum.
d- all of the above.

10-Incremental lines of Salter are:

a- highly mineralized areas having less collagen and great amount of mucopolysaccharides.
b- hypomineralized areas containing less collagen and great amount of mucopolysaccharides.
c- unmineralized areas containing no collagen nor mucopolysaccharides.
d- similar to cementum mineralization.

11-cementum hypertrophy occurs:

a- in non functioning teeth.
b- functioning teeth.
c- in embedded teeth.
d- in malposed tooth

12-Cementoblasts are derived from:

a- epithelial root sheath of Hertwig.
b- epithelial rests of Malassez.
c- cells of dental papilla.
d- ectomesenchymal cells of tooth sac.

13- Persistence of epithelial root sheath of Hertwig will result in:

a- Formation of intermediate cementum.
b- No cementum formation.
c- Formation of acellular cementum.
d- Formation of accessory root canal.

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Pulp

1- The nerve supply within the dental pulp provides for:

a- Pain perception, vascular control and proprioception.
b-Thermal perception, vasodilatation and cementum sensitivity.
c-Pain perception, dentin sensitivity and vascular control.
d-Proprioception, taste and vasoconstriction.

2- With age, the volume of the pulp decreases because dental pulp? of:

a- A decrease in the number of collagenous fibers.
b- The formation of secondary dentin.
c- Sclerosis of radicular dentin.
d- A decrease in number of fibroblasts.

3- The true pulp stone is formed usually:

a- In large size.
b- In the coronal pulp.
c- In the radicular pulp.
d- Around degenerated nidus.

4- The pulp sensation is:

a- Touch.
b- Propioceptive.
c- Pain sensation.
d- Cold.

5- As a result of aging, the dental pulp in human permanent teeth:

a- Increase in size and becomes more vascular.
b- Becomes less cellular and more fibrous.
c- Becomes more cellular and contains more ground substance.
d- Becomes more vascular and contains larger neural elements.

6- An embedded pulp stone is most likely to be surrounded by:

a- Fibrous tissue.
b- Primary dentin.
c- Secondary dentin.
d- Odontoblasts

7- Which of the following is thought to protect the dental pulp?

a- Sclerotic dentin.
b- Dead tracts.
c- Tome's granular layer.
d- Denticles

8- A false denticle:

a- Consists of dentinal tubules.
b- Is usually associated with a dead tract.
c- Indicates a dead or dying pulp organ.
d- Consists of concentric layers of calcification.

9- Dental pulp develops from:

a- Dental organ. b- Dental sac.
c- Dental papilla. d- Dental lamina.

10- Nucleus with cart wheel appearance is found in:

a- Plasma cells. b- Lymphocytes.
c- Progenitor cells. d- Mast cells.

11- Extracellular matrix of the connective tissue of the pulp is composed of:

a- Collagen fibers only. b- Ground substance only.
c- Both 1&2. d- None of the above.

12- Pericytes are contractile cells that surround the endothelial cells of:

a- Arterioles. b- Venules.
c- Lymph vessels. d- Capillaries.

13- Pulp tissue has receptors for:

a- Heat. b- Pain.
c- Pressure. d- All of the above.

6- The width of the periodontal ligament:

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Periodontal ligament

1- The PDL fibers running between two adjacent teeth:

a- Belong to the dento-alveolar group.
b- Are gingival fibers.
c- Are transseptal fibers.
d- Are oxytalan fibers.

2- Which of the following is not true of the alveolodental periodontal fibers? They:

a- Are organized into bundles with different orientation.
b- Are derived from epithelial rests of Malassez.
c- Undergo continuous remodeling.
d- Are formed and partially resorbed by fibroblasts.

3- Which of the following is not present in the periodontal ligament?

a- Epithelial rests of Malassez.
b- Fibroblasts.
c- Cementoblasts.
d- Osteocytes.

4- Which of the following principal fibers groups constitutes the main attachment of the tooth?

a- Oblique group.
b- Apical group.
c- Horizontal group.
d- Alveolar group.

5- Which of the following groups of tissues comprise the periodontium?

a- Alveolar bone, periodontal ligament, cementum.
b- Periodontal ligament, cementum, gingiva.
c- Cellular cementum, periodontal ligament, alveolar bone.
d- Cementum, alveolar bone, periodontal ligament, gingiva.

6- The periodontal ligament:

a- It is thiner when teeth are functioning normally.
b- It is narrowest at the level of the midroot area.
c- It becomes wider with aging.
d- It is of uniform width around multi-rooted teeth.

7- The periodontal ligament develops from:

a- Dental organ. b- Dental sac.
c- Dental papilla. d- Dental lamina.

8- Elasticity of periodontal ligament is due to:

a- Presence of elastic fibers.
b- Presence of oxytalan fibers.
c- Wavy coarse of collagen fibers.
d- None of the above.

9- The principal fibers of periodontal ligament are:

a- Elastic. b- Oxytalan.
c- Collagen. d- Precollagen.

10- Function of oxytalan fibers:

a- Support nerves.
b- Support blood vessels.
b- Help fibroblasts migration.
d- All of the above.

11- Open faced nucleus is found in:

a- Resorptive cells.
b- Synthetic cells.
c- Defensive cells.
d- Progenitor cells.

12- The periodontal ligament is formed of:

a- Cells only. b- Fibers only.
c- Ground substances only. d- All of the above.

13- The protective function of periodontal ligament is achieved by its:

a- Fibers.
b-Blood vessels.
b- Nerves.
d-All of the above.

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TOOTH DEVELOPMENT

1-Odontogenesis of the primary dentition begins at ..... wiu life.
2- The first stage of tooth development known as the .....
3- The vestibular lamina located ..... to the dental lamina.
4- An..... is the space between the maxilla and the mandible on one side and the lips and the checks on the other side
5-The second stage of odontogensis is called the .....
6-The ectomesenchymal cells condensation just beneath the enamel organ is called .....
7- The connective tissue-beneath-and around the enamel organ and dental papilla forming what is called .....
8- A ..... remains between the bud and the growing underlying ectomesenchyme.
9-The tooth primordium has three components ....., ... and dental sac
10-The dental organ is..... in origin, while both dental papilla and dental sac are ..... in origin.
11-The .....proposes that ectomesenchyme assumes the dominant role in crown pattern formation.
12- The .....proposes that epithelium assumes the dominant role in crown pattern formation.
13-The third stage of odontogenesis is called the .....
14- The stellate reticulum cells synthesize and secrete .....
15-The lingual extension of the dental lamina is called the .....
16-The .....are non succedaneous and have no primary predecessors.
17-The permanent molars develop from a .....of the dental lamina.
18-The fourth stage of odontogensis is called the .....
19-In the bell stage, the connection of dental organ to the dental lamina is elongated forming a .....
20-The point at which inner enamel epithelium differentiation first occurs represents the site of future .....
21-The region where the inner and outer enamel epithelia meet at the rim of the enamel organ is known as.....
22-The stellate reticulum layer is most fully developed at the .....
23- The cells of stratum intermedium contain much ..... enzyme which essential for enamel maturation.
24- The remnants of the dental lamina and the lateral dental lamina are called .....
25-After the crown of the tooth is formed, the dental papilla is called .....
26-The peripheral cells of the dental papilla will differentiate into .....whereas the inner cells are the primordium of the .....
27-The final stage of odontogensis is known as .....
28- The .....seen during the early stages of tooth development are enamel knot, enamel cord and enamel niche.
29- A layer of cells that seems to be essential to enamel formation but does not actually secrete the enamel is .....
30-The ..... is a strand of cells extending form the stratum intermedium into the outer enamel epithelium.
31-The process of root development takes place after the ..... is completely formed.
32-The..... is the structure responsible for root development.

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33-The function of ..... is to map out the root and induce dentin formation in the root area.
34- The remnants of epithelial root sheath of Hertwig are called .....
35- The number of roots that are formed is determined by the number of .....
36-..... is the union of the root structure of two or more teeth through the cementum only.
37- Concrescence commonly occurs in .....
38-Misplaced enamel formed on the cemental root surface is called .....
39- ..... is distortion in root (or roots) or crown angulations in a formed tooth.
40- .....is a deviation or bends restricted just to the root portion of the tooth and less than 90 degrees.
41-Supernumerary roots occurs mainly in the .....
42- The stimulus that initiates the actual formation of enamel matrix seems to be the presence of .....
43- The lateral dental lamina gives rise to .....
44- The epithelial rests in the periodontal ligament are derived from .....
45-The last hard dental tissue to be deposited is .....
46- The epithelial root sheath of Hertwig disintegrates ..... odontoblastic differentiation & dentin matrix deposition.
47- The dental lamina is induced to proliferate into a tooth bud by the .....
48- The ..... lies between inner enamel epithelium & stellate reticulum.
49- The epithelial component of the tooth germ is known as .....

ENAMEL

1- Ameloblasts having ruffled borders are..... ameloblasts.
2- ..... ameloblasts are characterized by having a distal cell membrane with many infoldings.
3-..... are a result of variation in rhythmic apposition.
4- Enamel.....are grass like structure.
5- Organic components of enamel are primarily removed from enamel by .....
6- Enamel is composed of rod,.....
7- The extensions from secretory ameloblast cells that give structure to enamel rods are called .....
8- .....*s* are due to rhythmic apposition of enamel.
9- ..... is formed of 96% minerals & 4% organic materials and water.
10- In enamel ..... appear to be periodic bands at 4 microns interval across the rod.
11- The complex arrangement of rods over the cusps of teeth is termed .....
12- Prismless enamel is .....mineralized than prismatic enamel.
13- The predominant inorganic portion of enamel is .....
14- Developmental enamel lamellae is type..... enamel lamellae.
15- The predominant organic portion of mature Enamel is.....
16-.....makes up the outside layer of anatomical crown of a tooth.
17- The ..... lines fissures, grooves and pits.
18- The property that enables enamel to withstand the mechanical forces applied during tooth functioning is .....
19- Enamel has a certain degree of ..... The main pathway is from saliva to the outer layer of enamel.
20- Mature enamel is composed of 88-90 % inorganic material and 10-12 % organic material and water by .....
21- Fluoro-apatite is 20% less soluble than ..... which means that it is much less susceptible to demineralization.
22- Because .....help killing cavity-causing bacteria. So, it used as topical application on the teeth to minimize caries.
23- 90% of the developing enamel proteins is a heterogeneous group of low-molecular weight hydrophobic proteins known as.....
24- .....secreted by differentiating ameloblasts.

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25- ..... interdigitate with the surface of the forming enamel giving it a picket fence appearance.

26- ..... determine the orientation of hydroxyapatite crystals and so responsible for the rod structure of enamel

27- ..... is a brief reorganization stage between enamel secretion and maturation.

28- During .....stage there is selective withdrawal of enamel proteins, mainly amelogenin, and The water. maturative ameloblasts are either .....ended (80 % of maturation ameloblasts) or

29 - .....ended (20 %).

30- In the .....ended ameloblasts the distal junctions are leaky and the proximal are tight.

31- The .....protects the enamel from being in contact with connective tissue cells in the dental sac. The reduced dental epithelium and the oral epithelium are joined to form the .....of the

32- erupting tooth.

33- Human enamel forms at a rate of ..... per day.

34- differentiating ameloblasts release small isolated drops of unmineralized enamel proteins called .....

35- .....is characterized by gradual completion of mineralization to reach the 96 % of the total weight of enamel.

36- During the maturation process,..... and ameloblastins are removed leaving ..... and tuftelin in the mature enamel.

37- Histological structure of..... could be described in ground sections only.

38- Under the cusp tip and incisal ridges, the course of the rods is more complicated where they become twisted and braided together and it is called .....

39-..... are an optical phenomenon, viewed by oblique reflected light.

40- .....overlying the cusps and incisal edges do not reach the surface unless there is some enamel loss due to attrition.

41- ..... mainly appears in deciduous teeth and first permanent molars and may be associated with increased caries susceptibility .

42- The quality of the .....enamel is better than that of the ..... enamel due to the more protected condition and constant nutrition.

43- ..... results from some odontoblastic processes, during the early stages of enamel formation, pushed themselves between the pre-ameloblasts.

44- Enamel ..... are most common beneath cusps where most crowding of odontoblasts would have occurred.

45- Enamel lamellae are..... or unmineralized and narrower, longer and less common than enamel tufts.

46- ..... lamellae are hypo mineralized. They never extend to dentin.They would contain more enamel proteins and less crystals than the rest of enamel.

47-..... is a crack occurs in the enamel before the tooth eruption due to a stimulus on the enamel surface causing its fracture.

48- ..... is a crack occurs in the enamel after the tooth eruption while it is functioning in the oral cavity.

*49-*The convexities of the scallops of the DEJ are directed towards the .....

50- Rodless enamel occurs as a result of the absence of ..... on the ameloblasts in the final stage of enamel formation.

51- Perikymata is numerous near the ..... and nearly disappear at the .....region.

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52- *53-*Rod-end markings are shallowest in the ..... and deepest near the .....surfaces.

54- .....cementum occurs in 60 % of cases.

contact . .....is the physiological wearing away of the tooth hard substance as a result of tooth to tooth

Dentin

1- ..... is the mineralized tissue that forms the bulk of the tooth.

*2-*Two major properties distinguish dentin from enamel: First, dentin is ..... Second, dentin .....

3- Dentin is slightly harder than.....or ..... and is less hard than .....

4- On radiographs, dentin appears more ..... than enamel and more ..... (lighter) than pulp.

6- The collagen of dentin comprises over 90% of the organic matrix. The principle collagen fibril is type ....., traces of type III and type V collagen are also found.

7-..... are highly specialized connective tissue cells that differentiate from the peripheral cellular layer of the dental papilla.

8- Secretory odontoblasts exhibit ..... and ..... activity along their plasma membranes.

9- The..... odontoblast produces dentin in a very slow rate.

10- In resting odontoblasts, there are increase in number and size of ..... and .....

11- The Mantle dentin varying in width from ..... µm. The large collagen fibrils (0.1-0.2 mm in diameter) aggregate in the cell free zone.

12- The collagen fibrils of mantle dentin are aligned ..... to the basal lamina, while in the mantle dentin of the root they are.....

13- .....theory proposes that fluid movement through the tubules distorts the local pulpal environment and is sensed by the free nerve endings in the plexus of Raschkow.

14- The .....dentin is the basic structure of dentin and forms its bulk.

15- The collagen fibers of ..... dentin are smaller in diameter (0.05Mm) and are more closely packed and interwoven with each other.

16- The fibers of circumpulpal dentin are present ..... to the tubules.

17- The ground substances of the ..... dentin are exclusively a product of odontoblasts.

18- The vesicles that bud from the odontoblast cells during mineralization of dentin are called .....

19- .....theory supposes that dentin contains nerve endings, which respond when dentin is stimulated.

20- The function of ..... is to provide a special micro-environment in which the first hydroxyappatite crystals can form.

21- the presence of matrix vesicles is limited to..... dentin.

22- ..... calcification founds in mantle dentin.

23-..... calcification usually presents in circumpulpal dentin formed just below mantle dentin.

24- In the rest of the circumpulpal dentin, a combined pattern of calcification occurs with a .....phase alternating with a ..... phase.

25- The convexity of the primary curvatures of dentinal tubules nearest the pulp chamber faces.....

26- The ..... result from the spiral track taken by the odontoblast during its course from the outer dentin surface to the pulp.

27- In some places of the crown, the peripheral ends of some odontoblastic processes cross the dentino-enamel junction and protrude into the enamel, forming .....

28- Dentin located between the dentinal tubules is called .....

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29- .....dentinistypeofhighlycalcifieddentinmatrixdepositedontheinternalsurfaceof the walls of the dentinaltubulesnarrowingthesizeofthelumen.

30- .....dentincontainslittleamountofcollagen.Itisabout40%moremineralizedthanintertubular dentin.

31- External stimulusaccelerates.....*r*dentinformationthateventuallymayobliteratethetubules spaces.

32- ..... dentinwillbeseenastranslucentringsingroundsections.

33-..... is the area of unmineralized or hypomineralized dentin where globular zones of mineralization have failedtofusewithinmaturedentin.

34- Inter-globular dentinisseenmostfrequentlyinthe.....dentinjustbelowthe.....dentin.

35- .....isaconstantfeatureoftherootdentinimmediatelyadjacenttocementum.

36- .....donotfollowanyincrementalpatternandusuallysmallerthaninterglobulardentin.

37-.....do not cross through Tomes' granular layer.

38- IncrementallinesofVon-Ebnerrunatrightanglesto.....

39- .....mayresultsfromacoincidenceofthesecondarycurvaturesbetweenneighboring dentinal tubules.

40- .....couldbeseeninalldeciduousteethaswellasinthemesiolingualcuspofthefirst permanent molar.

41- .....dentindevelopsbyodontoblastsafterrootformationhasbeencompleted.

42- The reparative dentinwithnotubulesatallnamed.....Whilethatwithtrappedformingcells named .....

43- .....ischaracterizedbycalcificationofdentinaltubules,wheretheodontoblasticprocesses undergo fatty degenerationandthencalcification.

44- .....dentinappearstranslucentwhenviewedbytransmittedlightwhilewithreflectedlightthese areas appear dark.

45- Dentin areas characterizedbydegeneratedodontoblasticprocesseshavebeencalled.....

46- Each .....issurroundedandisolatedbyanarrowzoneofscleroticdentin.

47- .....maybecausedinsometeethbythedeathofodontoblastscrowdedinnarrowpulpalhorns.

48- .....theorycontendsthatadentalstimulusexcitestheodontoblasticprocesswhich then transmits the excitationtotheadjacentnerveplexus.

Cementum

1-..... comprisesmatrixformationandmineralizationofcementum.

2- The .....willinducetheneighboringcellsofthedentalpapillatodifferentiateintoodontoblasts.

*3-*The epithelial rootsheathofHertwigwilllossitscontinuityanditscellsbecome.....

4- The cells entrappedinthemineralizedcementumarereferredtoas.....andoccupylacunae.

5- cellular cementumhasa.....mineralcontentthanacellularcementum.

6- The Sharpey's fibersof.....cementumgenerallyretainanunmineralizedcore.

7- The Sharpey's fibersof.....cementumexhibitacompletedegreeofmineralization.

8- Sharpey's fibers arederivedfrom.....,whicharenotcalcifiableintheiroriginallocation.

9- Although the .....islaiddownthroughoutlife,themineralcontentofthistissue,onceformed,does not change significantlywithage.

10- .....coversminorareasoftheenamel,particularlyatandalongthecementoenameljunction.

11- .....isuniqueinthatitisavascular,doesnotundergocontinuousremodelinglikebone,but continues to grow inthicknessthroughoutlife.

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12- ..... has significant function in tooth anchorage to the surrounding bone through numerous Sharpey's fibers inserting into it

13- The ..... can repair a resorptive defect of the root in a reasonable time due to its capacity to grow much faster than any other known cementum type.

14- The thickness of cementum at the cervical area is about .....and gradually increases in thickness till it reaches its maximum about ..... at the apex and at the bifurcation of the root.

15- .....contains 45% to 50% inorganic substances and 55% to 50% organic materials.

16- Under the light microscope two types of cementum can be differentiated: ..... and.....

17- The cementum is usually covered by a zone of ..... 3-4 microns wide in acellular cementum and is wider in cellular cementum.

18-..... covers the root dentin starting from the amelo-cemental junction till the apex, but it is often missing at the apical third of the

19- ..... have processes directed toward the periodontal surface of the cementum to provide nutrition.

20- It appears that the number of cells that become incorporated in cementum is proportional to the.....

21- In cementum, the .....are highly mineralized areas with less collagen and more groun substance.

*22-*The incremental lines of Salter are roughly ..... to the long axis of the root.

23- The cellular cementum is formed at ..... rate than the acellular cementum and the incremental lines ar therefore located further apart than in acellular cementum.

24- In ..... of all teeth, cementum meets the cervical end of enamel in a relatively sharp line.

25- In about .....of the teeth cementum overlaps the cervical end of enamel for a short distance.

26- Over compression of the periodontal ligament can result in the..... of cementum and dentin.

27- Cementum is markedly ..... on the distal than on the mesial root surface, indicating a relationship to mesial drift.

28- ..... is an abnormal thickening of cementum.

29-..... is increase of cementum in good function teeth to permit more periodontal fibers to be attached to the tooth.

30- Increase of cementum in a non-functioning tooth or in an embedded tooth, this is called .....

31- Cementum hyperplasia is characterized by the..... of Sharpey's fibers.

Periodontal ligament

1- The periodontal ligament is ..... that occupies the periodontal space between the root of the tooth and the alveolus

2- The width of PDL ranges from 0.15-0.38 mm, with the narrowest width at the ..... and its widest width at the..... followed by the apical region.

3- The PDL is formed from the ..... shortly after root development begins.

4- As the tooth erupts, the level of the alveolar crest comes to coincide with the .....and the oblique fiber bundles below the gingival fibers become horizontally aligned.

5- When the tooth finally comes into occlusion, the horizontally aligned fibers, termed....., become obliquely oriented.

6- The intercellular substances of PDL comprise .....

7- The connective tissue fibers are mainly collagenous. The main types of collagen in the PDL are types.....and .....

8- PDL fibers are divided into: .....&.....&.....

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9- Alveolo-dentalfibersaresubdividedintothefollowingfivegroups:

- .....
-.....
- .....
-.....
- .....

10- The trans-septalfibersgrouparelocatedjust.....onthemesialanddistal sides of the tooth only. The bundles of thesefibersrunfromthe.....over.....andinserted into the comparable region of the .....

11- The alveolarcrestgroupresiststhe.....movementsofthetooth.

12- The horizontalgroupresiststhe.....appliedtothetoothcrown.

13- The oblique grouprunningfromthecementuminan.....directiontobeinserted into the bone ..... This group prevents the apex of the root from being ..... against the bottom of the socket during.....pressure applied to the tooth crown.

14- The apical groupresistanyforcetendingto.....thetoothfromthesocket.

15- The inter-radicalgroupfunctionsin.....thetooth.

16- The accessory fibersrunfrombonetocementumin.....,moretangentially to..... of the tooth and found in the regionofthe.....group.

17- The oxytalan fibersconstitutenomorethan.....ofthePDLfibers.Theyareattached into the .....of the tooth and courseoutintothePDLinvariousdirections.

18- Oxytalan fibers mayhavearolein.....Italsosupportsthebloodvessels of the PDL during ..... and preventsthe.....ofthemundermasticatoryforces.

19- Cells of the PDLinclude:

.....
.....
.....
.....
.....

20- The epithelial restsofMalassezmayproliferatetoform.....ifstimulated.

21- The space betweenthecells,fibers,bloodvesselsandnervesinthePDLisoccupied by ..... It is made up of two major groupsofsubstances.....and.....

23- Periodontal nervefibersarebothmyelinatedandunmyelinated.Themyelinatedfibers are.....fibers only.

24- In human beingsthesmallnervefibersofPDLareresponsiblefor.....,while the large diameter nerve fibers terminate in specializedendingsknownas.....,responsiblefor.....

25- In PDL there arefourtypesofnerveendings:

-.....
-.....
- .....
-.....

26- The PDL throughagingshowsdecreasesin.....anddecreasesinthe..... It become ..... in thickness andmaycontain.....

27- The cementiclesinthePDLmaybe.....,.....or.....inthecementum, and are formed of ..... of the periodontal tissue.

28- FunctionsofPDLare:.....&……&……
&……..&……

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Answersof MCQ

Tooth development

d

1	c	31	d	61	b
2	d	32	d	62	d
3	c	33	a	63	a
4	a	34	a	64	b
5	d	35	d	65	c
6	c	36	a	66	d
7	d	37	a	67	d
8	c	38	a	68	d
9	d	39	b	69	b
10	d	40	b	70	a
11	d	41	c	71	a
12	b	42	a	72	b
13	d	43	a	73	c
14	a	44	b	74	d
15	d	45	a	75	d
16	a	46	a	76	a
17	c	47	b	77	b
18	d	48	a	78	d
19	b	49	c	79	d
20	d	50	d	80	c
21	b	51	c	81	c
22	c	52	b	82	b
23	d	53	d	83	a
24	a	54	a	84	a
25	d	55	c	85	b
26	a	56	d	86	d
27	a	57	a	87	c
28	a	58	d	88	b
29	b	59	b	89	a
30	d	60			

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Enamel

1 2	b	31	d
3 4	a	32	c
5 6	d	33	a
7 8	c	34	a
9	d	35	b
10	b	36	a
11	b	37	c
12	d	38	b
13	d	39	d
14	b	40	c
15	b	41	a
16	c	42	c
17	b	43	d
18	b	44	d
19	d	45	c
20	a	46	b
21	b	47	d
22	c	48	d
23	a	59	b
24	b	50	d
25	c	51	d
26	d	52	d
27	b	53	d
28	b	54	a
29	b	55	a
30	d	56	b
	c	57	b
	c	58	c
	b	59	c
	d	60	d

61	a	91	d
62	b	92	c
63	a	93	a
64	b	94	d
65	d	95	b
66	c	96	d
67	c	97	c
68	d	98	b
69	b	99	d
70	b	100	a
71	a	101	b
72	b		
73	b		
74	d		
75	b		
76	a		
77	c		
78	d		

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79	d		
80	b		
81	c		
82	c		
83	b		
84	a		
85	d		
86	b		
87	b		
88	d		
89	c		
90	c		

Dentin

Dentine

1	c	15	d
2	b	16	c
3	d	17	d
4	c	18	b
5	a	19	b
6	c	20	a
7	a	21	b
8	b	22	b
9	d	23	a
10	a	24	b
11	a	25	c
12	c	26	c
13	c	27	a
14	a	28	b

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Answersof MCQ

CEMENTUM

1	b	8 9	c
2	c	10	d
3	b	11	a
4	b	12	b
5	b	13	d
6	b		b
7	a		

PULP

1	c	8 9	d
2	b	10	c
3	c	11	a
4	c	12	c
5	b	13	d
6	c		b
7	a		

PERIODONTALLIGAMENT

1	c	8 9	c
2	b	10	c
3	d	11	d
4	a	12	b
5	d	13	d
6	b		d
7	b		

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Complete

TOOTH DEVELOPMENT

1- *sixth*
2- *initiation stage*
3- *buccal*
4- *oral vestibule*
5- *bud stage*
6- *dental papilla*
7- *dental sac (follicle)*
8- *basement membrane*
9- *enamel organ, dental papilla*
10- *ectodermal / ectomesenchymal*
11- *field model*
12- *clone model*
13- *cap stage*
14- *glucosaminoglycans*
15- *successional dental lamina*
16- *permanent molars*
17- *posterior extension*
18- *bell stage*
19- *lateral dental lamina*
20- *incisal edge or cusp*
21- *cervical loop*
22- *bell stage*
23- *alkaline phosphatase*
24- *epithelial rests of Serre's*
25- *dental pulp*
26- *odontoblasts / pulp*
27- *apposition stage*
28- *transitory structures*
29- *stratum intermedium*
30- *enamel cord*
31- *crown*
32- *cervical loop*
33- *root sheath*
34- *epithelial rests of Malassez*
35- *medial ingrowths at the epithelial diaphragm*
36-*Concrescence*
37- *permanent maxillary molars*

38- *enamel pearls*
39- *Dilaceration*
40- *Flexion*

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41- *permanent third molars*
42- *predendtin*
43- *deciduous teeth*
44- *epithelial root sheath of Hertwig's*
45- *cementum*
46- *after*
47- *ecto-mesenchyme*
48- *stratum intermedium*
49- *enamel organ*

ENAMEL

1- *maturative*
2- *Maturative*
3-*Incremental lines*
4- *tufts*
5- *maturative ameloblasts*
6- *interrod & rod sheath*
7- *Tomes processes*
8- *Striae of Retzius*
9- *Enamel*
10- *cross striations*
11- *Gnarled enamel*
12- *more*
13- *calcium*
14- *A*
15- *enamelin*
16-*Enamel*
17- *enamel*
18- *hardness*
19- *permeability*
20- *volume*
21- *hydroxyapatite*
22- *fluoride ions*
23- *amelogenins*
24-*Stippled material*
25-*Tomes' processes*
26- *Tomes' processes*
27- *Transitional stage*
28- *maturative*
29 - *ruffled/ smooth*
30- *smooth*
31- *reduced dental epithelium*
32- *dento-gingival junction*
~~33~~-6 *microns*
34- *stippled material.*
35- *Secondary maturation*
36- *amelogenins / enamelins*
37- *enamel*

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38- *gnarled enamel*
39-*Hunter-Schreger bands*
40- *Brown striae of Retzius*
41- *Neonatal line*
42- *prenatal / post natal*
43-*Enamel spindle*
44- *spindles*
45- *hypomineralized*
46- *Developmental*
47-*Type B lamella*
48- *Type C lamella*
49- *dentine*
50- *Tomes processes*
51- *cervical region / incisal or cusp*
52- *cervical region / occlusal and incisal*
53- *Afibrillar*
54- *Attrition*

Dentin

1- *Dentin*
2- *sensitive/ is formed throughout life.*
3- *bone / cementum / enamel.*
4- *radiolucent / radio-opaque*
6- *Icollagen*
7-*Theodontoblasts*
8- *alkaline-phosphatase / Ca-ATPase*
9- *resting*
10- *lysosomes / phagosomes.*
11- *20-150*
12- *atright angles / parallel to it.*
13- *Fluid or hydrodynamic*
14- *circumpulpal*
15- *circumpulpal*
16- *at right or oblique angles*
17- *circumpulpal*
18- "*matrix vesicles*"
19- *Directneural stimulation*
20- *matrix vesicles*
21- *mantle*
22-*Linear*
23-*Globular (calcospherite)*
24- *globular / linear*
25- *root ward*
26- "*secondary curvatures*"

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27- "*Enamel spindles*".
28- "*inter-tubular dentin*"
29- *Peri-tubular "intratubular"*
30- *Intratubular*
31- *peritubular*
32- *Peritubular*
33-*Inter-globular dentin*
34- *circumpulpal / mantle*
35- *Tomes' granular layer*
36- *Tomes' granular layer*
37-*Thedentinal tubules*
38- *thedentinal tubules*
39- *Contour line of Owen*
40- *Neonatal line*
41-*Regular secondary*
42- "*atubular dentin*"/ "*osteodentin*"
43- *Sclerotic dentin*
44- *Sclerotic*
45- "*dead tracts*"
46- *dead tract*
47-*Dead tracts*
48- *Odontoblastic transduction*

Cementum

1-*Cementogenesis*
2- *inner dental epithelium*
3- *the epithelial rests of Malassez.*
4- *cementocytes*
5- *lower*
6- *cellular*
7- *acellular*
8- *periodontal fibers*
9- *cementum*
10-*Acellular afibrillar cementum*
11- *Cementum*
12-*Acellular extrinsic fiber cementum*
13- CIFC
14- *20-50 microns / [150-200u]*
15-*Cementum*
16- *the acellular/ the cellular cementum.*
17- *cementoid tissue*
18-*Acellular cementum*
19-*Cementocytes*
20- *rate of cementum deposition.*
21- *incremental Lines of Salter*
22- *parallel*

HMHDental

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23- *a faster*
24- *30%*
25- *60%*
26- *resorption*
27- *thicker*
28-*Hypercementosis*
29-*Cementum hypertrophy*
30- *cementum hyperplasia*.
31- *absence*

Periodontal ligament

1- the dense fibrous connective tissue
2- middle third of the root
3- dental follicle
4- cementoenamel junction
5- alveolar crest group,
6- fibers and ground substances
7- I / III
8- The principal fibers / The accessory fibers / The oxytalan fibers.
9- Alveolar crest group / Horizontal group / Oblique group / Apical group / Inter-radicular group
10- apical to the gingival group / cementum of one tooth / the alveolar crest / cementum of the adjacent tooth.
11- horizontal
12- horizontal pressure
13- oblique / coronally. / jammed / vertical
14- lift
15- stabilizing
16- different planes / prevent rotation / horizontal
17- 3% / cementum
18- tooth support / mastication / sudden closure
19- Synthetic cells: (fibroblasts, osteoblasts and cementoblasts) / Resorptive cells: (cementoclasts, osteoclasts and fibroclasts) / Progenitor cells: (undifferentiated mesenchymal cells) / Defensive cells: (macrophage, lymphocytes and mast cells) / Epithelial cells: (epithelial rests of Malassez).
20- cysts or tumors
21- ground substance / proteoglycans /glycoproteins.
22- superior / inferior alveolar / Branches from the apical vessels that supply the dental pulp / Branches from the intra-alveolar vessels / Branches from the gingival vessels.
23- sensory
24- pain sensation / mechanoreceptors / localization of touch and pressure on the teeth.
25- free nerve endings / Ruffini's corpuscles / coiled form / spindle-like ending.
26- vascularity / number of its cells / thinner / cementicles.
27- free / attached / embedded / dystrophic calcification.
28- Supportive / Sensory / Nutritive / Formative / Protective

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Write the scientific term:

TOOTH DEVELOPMENT

1- It contributes the development of the vestibule of the mouth.
2-The Lingual extension of the dental lamina
3- The first stage of tooth development.
4- The space between the maxilla and the mandible on one side and the lips and the checks on the other side
5- The second stage of odontogensis.
6-The ectomesenchymal cells condensed just beneath the enamel organ
7- The connective tissue-beneath-and around the enamel organ and dental papilla.
8-The third stage of odontogensis.
9- The region where the inner and outer enamel epithelia meet at the rim of the enamel organ.
10-The enzyme that secreted by cells of the stratum intermedium and essential for enamel maturation.
11- The remnants of the dental lamina and the lateral dental lamina.
12- The final stage of odontogensis.
13- A layer of cells that seems to be essential to enamel formation but does not actually secrete the enamel.
14-The strand of cells extend form the stratum intermedium into the outer enamel epithelium.
15-The remnants of epithelial root sheath of Hertwig.
16- The union of the root structure of two or more teeth through the cementum only.
17-Misplaced enamel formed on the cemental root surface.
18- The distortion in root (or roots) or crown angulations in a formed tooth.
19- The deviation or bends restricted just to the root portion of the tooth and less than 90 degrees.
20- The stimulus that initiates the actual formation of enamel matrix.
21- The downgrowth of an epithelial thickening buccal to the dental lamina.
22- The epithelial component of the tooth germ.

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Answers of Tooth Development

1- Vestibular lamina
2- Successional dental lamina
3-The initiation stage
4-An oral vestibule

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5-The bud stage. 6- Dental papilla. 7- dental sac (follicle). 8- The cap stage. 9- The cervical loop

10- Alkaline phosphatase
11-Epithelial pearls.
12- Apposition stage.
13- Stratum intermedium
14-Enamel cord
15- Epithelial rests of Malassez.
16-Concrescence
17-Enamel pearls.
18-Dilaceration
19- Flexion
20- Predentin
21- Vestibular lamina.
22-Enamel organ

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ENAMEL

1-Is the physiological wearing away of the tooth hard substance as a result of tooth to tooth contact.
2- It is the predominant organic portion of mature enamel.
3-The dental tissue that lines fissures, grooves and pits.
4-The property that enables enamel to withstand the mechanical forces applied during tooth functioning.
5- The ions that help killing cavity-causing bacteria. So, it used as topical application on the teeth to minimize caries.
6- The structures that interdigitate with the surface of the forming enamel giving it a picket fence appearance.
7- It determines the orientation of hydroxyapatite crystals and so responsible for the rod structure of enamel.
8- It is a brief reorganization stage between enamel secretion and maturation.
9- It protects the enamel from being in contact with connective tissue cells in the dental sac.
10- The small isolated drops of unmineralized enamel proteins that are released by differentiating ameloblasts.

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11- The process that is characterized by gradual completion of mineralization to reach the 96 % of the total weight of enamel.
12-The enamel under the cusp tip in which the course of the rods is more complicated and braided together. .
13- The line that appears in deciduous teeth and first permanent molars and may be associated with increased caries susceptibility .
14- It results from some odontoblastic processes, during the early stages of enamel formation, pushed themselves between the pre-ameloblasts.
15-Lamella that is hypo mineralized and never extends to dentin.
16- It is a crack occurs in the enamel before the tooth eruption due to a stimulus on the enamel surface causing its fracture.
17- It is a crack occurs in the enamel after the tooth eruption while it is functioning in the oral cavity.

Answers of Enamel

1-Attrition
2-Enamelin.
3-The enamel
4-Hardness.
5-Fluoride
6-Tomes' processes
7-Tomes' processes
8-Transitional stage
9-The reduced dental epithelium
10-Stippled material.
11-Secondary maturation
12-gnarled enamel
13-Neonatal line
14-Enamel spindle.
15-Developmental
16-Type B lamella
17-Type C lamella

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DENTIN

1- The mineralized tissue that forms the bulk of the tooth.
2- The mineralized tissue which is slightly harder than cementum and is less hard than enamel.
3-The highly specialized connective tissue cells that differentiate from the peripheral cellular layer of the dental papilla.
4- The theory supposes that dentin contains nerve endings, which respond when dentin is stimulated.
5- The vesicles that bud from the odontoblast cells during mineralization of dentin.
6-Its function is to provide a special micro-environment in which the first hydroxyappatite crystals can form.
7- Type of calcification usually presents in circumpulpal dentin formed just below mantle dentin.
8-It results from the spiral track taken by the odontoblast during its course from the outer dentin surface to the pulp.
9-Structures that result from the peripheral ends of some odontoblastic processes crossed the dentino-enamel junction and protrude into the enamel.
10-type of dentin located between the dentinal tubules.
11- Type of highly calcified dentin matrix deposited on the internal surface of the walls of the dentinal tubules narrowing the size of the lumen.
12- The area of unmineralized or hypomineralized dentin where globular zones of mineralization have failed to fuse within mature dentin.
13- A constant feature of the root dentin immediately adjacent to cementum.
14- Lines that may results from a coincidence of the secondary curvatures between neighboring dentinal tubules.
15- Structure could be seen in all deciduous teeth as well as in the first permanent molar.
16-Type of dentin develops by odontoblasts after root formation has been completed.
17- The type of dentin with no tubules at all.
18- The type of dentin characterized by calcification of dentinal tubules, where the odontoblastic processes undergo fatty degeneration and then calcification.
19-Dentin areas characterized by degeneration of their odontoblastic processes.
20-The type of dentin which is surrounded and isolated by a narrow zone of sclerotic dentin.
21- The theory contends that a dental stimulus excites the odontoblastic process which then transmits the excitation to the adjacent nerve plexus.
22- The theory proposes that fluid movement through the tubules distorts the local pulpal environment and is sensed by the free nerve endings in the plexus of Raschkow.

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23- The type of dentin which showed trapped forming cells.

Answersof Dentin

1-Dentin 2-Dentin 3-Odontoblasts 4-Direct neural stimulation 5- Matrix vesicles 6- Matrix vesicles 7- Globular calcification (calcospherite)
8- The "secondary curvatures"
9- Enamel spindles
10- Inter-tubular dentin.
11- Peri-tubular or "intratubular" dentin
12- Inter-globular dentin.
13- Tomes' granular layer.
14- Contour line of Owen
15- Neonatal line
16- Regular secondary
17- Atubular dentin
18- Sclerotic dentin
19- Dead tracts. 20- Dead tract 21-Odontoblastic transduction theory 22-Fluid or hydrodynamic

23- Osteodentin

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Cementum

1-It comprises matrix formation and mineralization of cementum.
2- The cell layer of dental organ which induce the neighboring cells of the dental papilla to differentiate into odontoblasts.
3-The cells that remain after loss of the continuity of the epithelial root sheath of Hertwig..
4- The cells entrapped in the mineralized cementum and occupy lacunae.
5-It consists of a mineralized matrix, which contains neither collagen fibrils nor embedded cells and covering the cervical portion of enamel.

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6-They are derived from periodontal fibers, which are not calcifiable in their original location but they will calcify after they have become embedded in bone and cementum only.
7-Its formation starts at the end of enamel maturation when the cells of the reduced enamel epithelium be lost or detached from the enamel.
8- Areas of human teeth are covered with thick cementum layers before they emerge into the oral cavity.
9- Mineralized tissue is unique in that it is avascular, does not undergo continuous remodeling like bone, but continues to grow in thickness throughout life.
10-Dental tissue contains 45% to 50% inorganic substances and 55% to 50% organic materials.
11- It covers the root dentin starting from the amelo-cemental junction till the apex, but it is often missing at the apical third.
12- Cells have processes directed toward the periodontal surface to provide nutrition.
13-Incremental Lines are highly mineralized with less collagen and more ground substance.
14-It is an abnormal thickening of cementum.
15- Increase of cementum in good function teeth to permit more periodontal fibers to be attached to the tooth.
16-Iincrease of cementum in a non-functioning tooth or in an embedded tooth.

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Answers of Cementum

1-Cementogenesis
2-Inner dental epithelium
3-The epithelial rests of Malassez
4-Cementocytes
5-Acellular afibrillar cementum
6- Sharpey's fibers
7-Acellular afibrillar cementum
8- The furcations
9- Cementum
10-Cementum
11- Acellular cementum
12- cementocytes
13- Incremental Lines of Salter
14-Hypercementosis
15- Cementum hypertrophy
16- Cementum hyperplasia

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1-Cementum may cover part of the enamel due to:
a) Premature destructionof the epithelialrootsheathof Hertwig.
B) Premature differentiation of odontoblast
C) Malfunctioning of the maturative ameloblast
D) Destruction of the reduced enamel epithelium covering this part

2-In severely stimulated tooth, reparative dentin is laid down by:
A)Newlydifferentiatedodontoblasts.
B)Both newly differentiated and old survived odontoblasts
C)Old odontoblasts survived after injury
D)Old ameloblast survived after injury

3-The following statement is/are true about DEJ.
A)DEJ appear as scalloped line
B)The surface of dentin at DEJ is pitted.
C)The surface of dentin at DEJ is smooth.
D)This relation assumed the firm hold of the enamel cap ondentinsurface.

4-Odontoblastic process which passes across the DEJ into enamel is called:
A)Enamel spindle.
B)Gnarled enamel.
C)Enamel lamellae.
D)Enamel tuft.

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5-Which of the following is not a feature of cellular cementum?
A)Incremental lines.
B)Sharpey's fibers.
C)Continuous remodeling
D)Lacunae and canaliculi

6-Cementum may be deposited on the surface of enamel as a consequence of:
A)Injury to thereduced enamelepithelium
B)During passive eruption
C)Early degeneration of Hertwig's sheath.

7-Cementoblasts:
A)Are multinucleated cells.
B)May become incorporated into acellular cementum.
C)Are derived from the dental papilla.
D)Secrete both collagen and glycosaminoglycans.

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8-Odontoblast cell bodies are present in the:
A)Cementum.
B)Dentin.
C)Pulp
D)Enamel.

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9-The dentin is formed:
A)Slightly after periodontal ligament.
B)Slightly after the enamel.
C)Slightly after the cementum.
D)Slightly before theenamel.

10-Which of the following is true about cementum? It
A)Provides forthe attachment of principalperiodontal fibers.
B)Are more easily resorbedthan bone.

11-When odontoblasts recede towards the pulp they leave a part of their cytoplasm termed?
A)Von Korff's fibers.
B)Von Ebner incremental lines of dentin.
C)Tome's fiber with the formed dentin.
D)Tome's process of ameoblast.

12-The highly calcified dentin is:
A)Interglobular dentin.
B)Tome's granular layer.
C)Intertubular dentin.
D)Peritubular dentin.

13-Dentin in which the tubules are calcified is termed as:
A)Sclerotic dentin.
B)Interglobular dentin.
C)Dead tract dentin.
D)Physiological secondarydentin

14-Which is considered right about cellular cementum:
A)Located atcervical2/3 of root.
B)First formed cementum.
C)Formed after acellular cementum.
D)Don't have cementocyte cells.

15-The thickness of acellular cementum:
A)150-200 m
B)30-40 m
C)20-50 m
D)60-80 m

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17-Most peripheral part of primary dentin is:
A)Predentin
B)Mantle dentin
C)Intertubular dentin
D)Circumpulpardentin

18-The following dental tissue is laid down first:
A)Enamel
B)Dentin
C)Pulp
D)Cementum

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19-Perikymata are:
A)Verticallyoriented wave like grooves.
B)Parallel tothe AEJ.
C)Numerousat the incisal or cusp regions.
D)Theexternal manifestation ofthebrown striaeofRetzius.
20-In an oblique plane, the bundles of rods seem to interwine more irregularly in the regions of cusps or incisal edge, this appearance of enamel is called: A)Gnarledenamel. B)Hunter-Schreger band. C)Striations. D)Incrementallines ofRetzius. **21-The thickness of prismless enamel layer in primary teeth is** A)75 microns B)100 microns C)30 microns D)50 microns **22-The epithelial root sheath of Hertwig's is composed of:** A)Inner enamel epithelium & stratumintermedium. B)Stellate reticulum & stratum intermedium. C)Outer enamel epithelium & Inner enamel epithelium

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D)Outer enamel epithelium & stellatereticulum
23-The morphological outline of the crown of teeth is controlled by:
A)Home box genes
B)Induction by odontoblasts cells.
C)Growth factors.
D)Nutrition & vitamins.
24-The successor of deciduous teeth develops from:
A)Occlusal extension of the dental lamina.
B)Buccal extension of the dental lamina.
C)Lingual extension of the dental lamina.
D)Labial extension of the dental lamina.
25-The final product of embryonic dental papilla is:
A)Pulp, dentin, cementum & periodontal ligament.
B)Pulp, dentin & cementum only.
C)Dentin & cementum only.
D)Pulp & dentin only.

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26-If cells of the epithelial root sheath remain adherent to dentin surface, they may differentiate into ameloblasts and produce enamel. Such droplets of enamel are called:
A)Epithelialdiaphragm.
B)Rest of Mallasez
C)Enamel pearls.
D)Epithelialpearls.
27-In transition stage of enamel formation, the following changes take place:
A)Secretion of enamel begins.
B)Increase length of ameloblast.
C)Secretion of enamel just started.
D)Removal of amelogenin starts.
28-Incremental lines of enamel are a result of variation in:
A)Morphodifferentiation.
B)Proliferation
C)Histodifferentiation
D)Rhythmic apposition.
29-Development of accessory root canal is due to:
A)The broken continuity of Hertwig's root sheath.
B)All of above
C)The apposition of dentin and cementum to the apex of the root.
D)Development of tongue-like extension of the horizontal diaphragm
30-The shape and size of teeth is determined by which stage:
A)Morphodifferentiation
B)Histodifferentiation
C)Initiation
D)Apposition
31-The epithelial root sheath of Hertwig helps to:
A)Form the radicular
B)Form crown of tooth
C)Form the radicular dentin
D)Mold the shape of roots.
32-The cells of dental sac will form:
A)Dentin
B)Pulp
C)Enamel
D)Cementum

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33-Hunter-Schreger bands are:
A)Run parallel to stria of Retzius.
B)Reflects periodic changes in prism direction of enamel
C)Found in the outer surface of enamel
D)Best visualized in transverse section of the crown

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34-In transverse sections of tooth, the incremental lines of Retzius appears as:
A)Oblique line
b)Concentric circle.
C)Crack
D)Zigzag line.

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35-The DEJ is in the form of:
A)Zigzag line
B)Straight line
C)Scalloped with convexities of scallops toward dentin
D)Scalloped with convexities of scallops toward enamel

36-The enamel cord:
A)Is an extension of the dental lamina
B)Is an extension of successional dentallamina
C)Is an extension of the inner dentalepithelium
D)Is an extension of the outer dentalepithelium

37-Direction of enamel rods in deciduous and permanent teeth is:
A)Different at middle third
B)Different at cervical third
C)Different at occlusal third
D)Same throughout

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38-Thin, leaf-like structures that extends from enamel surface towards the DEJ are called:
A)Gnarled enamel
B)Enamel lamella
C)Enamel tufts
D)enamel spindle

39-The shape of the root is determined by cells of the:
A)Dental organ
B)Vestibular lamina
C)Dental sac
D)Dental papilla

40-In the dental organ, the stratum intermedium cells
A)Play no role in enamel calcification
B)Are highly gelatinous and rich in mucopolysaccharides.
C)Lie Between inner enamel epithelium & stellate reticulum
D)Don't provide a reserve source of cells for the stratum intermedium.

41-In three-rooted teeth:
A)The diaphragm surrounds three cervical openings
B)The diaphragm surrounds four cervical openings
C)Four tongue like projection grow and fuse
D)Three tongue like projection grow and fuse

42-Regraded to formation of enamel and primary dentin in developing tooth, the following sentence is true:
A)The enamel is formed more cervically thandentin
B)The ameloblasts & odontoblasts move toward pulp
C)The enamel is formed slower than dentin
D)The enamel is formed faster than dentin

43-The droplets of enamel found in furcation area of the roots of permanent molars are:
A)Enamel organ
B)Enamel pearl
C)Enamel spindle
D)Enamel cracks

44-The remnants of epithelial root sheath found in periodontal ligament are called
A)Enamel knot
B)Enamel pearl
C)Epithelial diaphragm
D)Restsof Mallasez

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M.C.Q. QUESTION

Tooth development

1-Concrescence is:

a- An extra root or accessory roots in a formed tooth.
b- Distorted root or roots in a formed tooth.
c- A union of root structure of two or more teeth through cementum only.
d- A spherical projection on the cemental root surface.

2- What structure is formed in the crown first?

a-Enamel.
b-Cementum.
c-Pulp.
d- Dentine.

3- Which of the following induces the dental papilla cells to be differentiated into odontoblasts?

a-Stratum intermedium. b-Reduced enamel epithelium. c-Inner enamel epithelium. d-Outer enamel epithelium.

4- No basal lamina is found between the cells of the:

a-Stratumintermediumand inner enamel epithelium.
b-Outer enamel epithelium and the dental sac.
c-Inner layer of Hertwig's sheath and the dental papilla.
d- Inner enamel epithelium and the dental papilla.

5- Which of the following is <u>not</u> a functional activity of the enamel organ:

a-Inducing the differentiation of Odontoblasts.
b-Secretion of enamel matrix.
c-Maturation of enamel.
d- Formation of cementum.

6- Which of the following is <u>not</u> derived from the dental organ:

a- Stellatereticulum.
b-Hertwig's epithelial root sheath.
c- Odontoblasts.
d- Ameloblasts.

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7- The stratum intermedium:

a- Inducesdentinformation.
b-Isseparatedfromthestellate reticulum by a basal lamina.
c- Develops incapstage.
d- Isimportant for enamel maturation.

8- Thedentalorgan

a-Developsfromcellsinthe dental follicle.
b-Isacompletelyconnective tissue structure.
c-Isacompletelyepithelial structure.
d-Ishighlyvascular,asameloblasts require an enriched environment.

9- Theenamel cord

a-Isanextensionofthedental lamina.
b-Isanextensionoftheouter dental epithelium.
c-Isanextensionofsuccessional dental lamina
d-Extendsfromthestratum intermedium to the stellate reticulum.

10-Dental folliclecells

a-Differentiateintotheameloblast layer.
b-Differentiateintotheodontoblast layer.
c-Arefoundwithinthedental organ.
d-Migratetodentinsurfaceof the rootanddifferentiateintothecementoblast.

11- All of thefollowingis thefunctionsof TheStellate Reticulumexcept:

a-KeepSpaceforTheenamel development.
b-Actasareservoirfornutritive materials.
c-Actasacushionedtoprotect ameloblasts.
d-Secretesalkalinephosphatase enzyme .

12- The epithelial root sheath of Hertwig disintegrates:

a-Afterodontoblasticdifferentiation & before dentin matrix formation.
b-Afterodontoblasticdifferentiation & dentin matrix deposition.
c- Beforeodontoblasticdifferentiation.
d- After cementum formation.

13- All of the following is a function of dental sac except:

a- Nutritionof enamel organ.
b- Formationof cementum.
c-Formationofperiodontal ligament.
d-Formationofsupporting alveolar bone.

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14- The lateral dental lamina gives rise to:

a- Deciduous teeth.
b- Permanent successors.
c- Permanent molars. d- Vestibular lamina.

15- The stratum intermedium:

a- Lies between outer enamel epithelium & stellate reticulum.
b- Is a transient structure.
c- Is rich in mucopolysaccharides.
d- Plays important role in enamel calcification.

16- The down growth of an epithelial thickening buccal to the dental lamina is known as:

a- Vestibularlamina.
b- Linguo-alveolar Sulcus.
c- Lateral dental lamina.
d- Successional dental lamina.

17- The dental lamina is induced to proliferate into a tooth bud by the:

a- Basementmembrane.
b- Nerve endings.
c- Ecto-mesenchyme.
d- Oral epithelium.

18- Calcified tissues of the tooth are derived from:

a- Ectoderm only. b- Endoderm only. c- Mesoderm only. d- Ectoderm & Mesoderm.

19- A layer of cells that seems to be essential to enamel formation but does <u>not</u> actually secrete the enamel is:

a- Outerdental epithelium.
b- Stratum intermedium.
c- Reduced dental epithelium.
d- Vestibular lamina.

20- The stimulus that initiates the actual formation of enamel matrix seems to be:

a- Reduction of stellate reticulum.
b- Influence of increased vascularization of the dental sac.
c- Disappearance of the cell free zone of dental papilla.
d- Presence of predentin.

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21- The number of roots that are formed is determined by the:

a-Number of root sheaths developed by the enamel organ.
b-Number of medial ingrowths at the epithelial diaphragm.
c-Number of root sheaths developed by the dental sac.
d-Thickness of the cervical loop.

22- Which of the following is the first process to occur in the sequence of tooth development:

a-Deposition of the first layer of enamel.
b-Deposition of the first layer of dentin.
c-Elongation of the inner dental epithelial cells.
d-Differentiation of odontoblasts.

23-All of the following are involved in the formation of a tooth <u>except</u>:

a-Epithelial root sheath.
b-Successional lamina.
c-Dental lamina.
d-Vestibular lamina.

24- The dental lamina initiating the permanent molars develops:

a-As successional lamina.
b-As a distal extension of the dental lamina.
c-As lateral dental lamina.
d-As vestibular lamina.

25-The cell rests of Malassez are derivatives of:

a-Cervical ameloblasts.
b-Outer enamel epithelium.
c-Dental papilla.
d-Root sheath.

26-Odontogenesis of the primary dentition begins between:

a- Thesixth & seventh week. b- Thefifth & sixth week. c- Theseventh& eighth week. d- Thefourth & fifth week. **27- The embryo's stomodeum is lined by:** a- Ectoderm. b- Endoderm. c- Mesoderm. d- Ectomesenchyme.

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28- By the 6th week of tooth development, the oral epithelium is:

a- 2-3 layers thickness.
b- 3-4 layers thickness.
c- 4-5 layers thickness.
d- 2-5 layers thickness.

29- The vestibular lamina gives rise to:

a- The alveolodental sulcus.
b- The alveolobuccal sulcus.
c- The alveololingual sulcus.
d- The alveologingival sulcus.

30- The ectomesenchymal cell condensation just beneath the enamel organ is called:

a- Dental sac.
b- Dental follicle.
c- pulp.
d- Dental papilla.

31- The cells of the tooth bud have:

a- High RNA content only.
b- Low glycogen content only.
c- Increased oxidative enzyme activity only.
d- All of the above.

32- Regulatory genes play a significant role in tooth development, they:

a- Control the tooth position.
b- Control the tooth number.
c- Determine the tooth shape.
d- a & b.

33- Initiation of dental lamina is induced by:

a- Neuralcrestcell.
b- Endoderm.
c- Ectoderm.
d- Mesoderm.

34- Primordium for the permanent dentition appears as an extension of dental lamina into the ectomesenchyme:

a- Lingual to the developing primary tooth germ.
b- Labial to the developing primary tooth germ.
c- Mesial to the developing primary tooth germ.
d- Distal to the developing primary tooth germ.

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35- The region where the inner and outer enamel epithelium meets at the rim of the enamel organ is known as:

a- Zone of Flection.
b- Cervical loop.
c- Epithelial Diaphragm.
d- Root Sheath of Hertwing's.

36- Which the followings *is not* _____a function of stellate reticulum:

a- Protection of the underlying dental tissues. b- Maintenance of tooth shape. c- Support the production of enamel. d- Control the position & number of tooth germs. **37- The epithelial pearls are derived from:** a- Dental lamina. b- Dental pulp. c- Dental sac. d- Dental organ. **38- Odontoblasts start their secretory activity:** a- Before enamel matrix production. b- After enamel matrix production. c- During enamel matrix production. d- During & after enamel matrix production. **39- Preameloblasts differentiate into ameloblasts:** a- After the differentiation of odontoblasts. b- Before the differentiation of odontoblasts. c- During the differentiation of odontoblasts. d- During & after the differentiation of odontoblasts.

40- During the early stages of tooth development three transitory structures may be seen, these structures are:

a- Enamel knot, enamel organ & enamel niche.
b- Enamel knot, enamel cord & enamel niche.
c- Enamel knot, enamel organ & enamel epithelium.
d- Enamel niche, enamel organ & enamel knot.

41- The epithelial rests in the periodontal ligament are derived from:

a- Dental pulp.
b- Epithelial root sheath of Hertwing's.
c- Vestibular lamina.
d- Dental sac.

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42- Appositional stage is confined to:

a- Cap stage. b- Bud stage.
c- Bell stage. d- Dental lamina.

43- Concresense usually occurs with: a- Permanent maxillary molars. b- Permanent maxillary premolars. c- Permanent mandibular premolars. d- Permanent mandibular premolars.

44- Supernumerary roots occur mainly with: a- Permanent third molars. b- Permanent second molars. c- Permanent first molars. d- Permanent second premolars. **45- Enamel pearls occur in:** a- Apical third of molars roots. b- Furcation area of molars. c- Cervical third of molars roots. d- Furcation area of premolars.

46- The process of differentiation of mesenchymal cells into Odontoblasts is called:

a-Initiation. b- Induction.
c-Apposition. d- Morphodifferentiation.

47- The epithelial root sheath of Hertwing's is composed of:

a-Outer enamel epithelium & inner enamel epithelium.
b- Inner enamel epithelium & stratum intermedium.
c- Outer enamel epithelium & stellate reticulum.
d- Stellate reticulum & stratum intermedium.

48- Inner dental epithelium:

a- Consists of a single layer of cuboidal epithelial cells.
b- Consists of a single layer of columnar epithelial cells.
c- Consists of a single layer of polyhedral epithelial cells.
d- Consists of a single layer of polygonal epithelial cells.

49- Determination of the initial position of the first cusp of the tooth during crown pattern formation may be achieved by:

a- The enamel knot. b- Theenamel cord.
c- Both enamel knot andcord. d- None of the above.

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50- Before enamel formation begins:
a- Enamel knot disappearsandenamelcord remains.
b- Enamel knot remains and enamel cord disappear.
c- Both knot and cord disappear.
d- Both knot and cord remain.

51- Concerning the stellate reticulum one of the following statements is wrong:
a- Occupies the central portion ofthe dentalorgan.
b- Lies between outer and inner dental epithelia.
c- Synthesizes and secretes glucoseaminoglycans.
d- Becomes polyhedral in shape.

52- The dental sac is:
a- The condensation of the ectomesenchymal cells surrounding the dental organ.
b- The condensation of the ectomesenchymal cells surrounding the dental papilla.
c- The condensation of the ectomesenchymal cells surrounding the dental organ and dental papilla.
d- None of the above.

53- The main difference between bell and apposition stage is:
a- The formation of first layer of E.
b- The formation of first layer of D.
c- The formation of first layer of C.
d- The formation of first layer alveolar bone.

54- Concerning the stratum intermedium, one of the following statements is incorrect:
a- It is composed of 2-3 layers of squamous epithelialcells.
b- It is differentiated between the inner enamel epithelium and stellate reticulum.
c- It is connected to each other and to the cells of stratum intermedium by desmosomes.
d- It is essential for enamel formation as it contains acid phosphatase enzyme.

55- Which one of the following functions of the inner dental epithelium is wrong?
a- It exerts an organizing influence on the undifferentiated cellsofthe dental papilla to differentiate into ameloblasts.
b- It shares in the transport of the nutritive materials.
c- It determines the future incisal or occlusal crown pattern.
d- It lays down enamel matrix and helps in its mineralization.

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56- The epithelial root sheath of Hertwig can't be seen as a continuous layer in the developing root because:
a- The rapid epithelial sheath proliferation.
b- The dentin formation.
c- The rapid epithelial sheath destruction after dentin formation.
d- All of the above.
57- Epithelial rests of Malassez are:
a- Epithelial strands and may undergo degeneration only.
b- Epithelial remnants of the epithelial root sheath of Hertwig only.
c- Network or isolated islands in the P.L. only,
d- All of the above.
58- If Hertwig's epithelial root sheath does not disintegrate:
a- No cementum will be formed on the radicular dentin.
b- It is likely that some of the enamel will be resorbed.
c- The apical end of the junctional epithelium will lie occlusal to the cemento-enamel junction.
d- Predentine will form on the outer surface of the radicular dentin.
59-The reduced enamel epithelium is derived from:
a- Maturative ameloblasts and stratum intermedium.
b- Stratum intermedium, stellate reticulum and outer dental epithelium.
c- Stratum intermedium, stellate reticulum and inner epithelium.
d- Protective ameloblasts, stratum intermedium, stellate reticulum and outer dental epithelium.
60-Which of the following is not component of the enamel organ?
a- Ameloblasts.
b- Odontoblasts.
c- Outer enamel epithelium.
d- Stratum intermedium.

61- Which of the following indicates the major components of a tooth germ?
a- The dental lamina, the stellate reticulum and the stratum intermedium.
b- The dental organ, the dental papilla and the dental sac.
c- The dental pulp, the dental sac and the dental lamina.
d- The cervical loop, Hertwig's sheath and the inner enamel epithelium.
62- Which of the following is not ectodermal in origin?

a-Epithelial rests of Malassez.
b-Anterior pituitary gland.
c-Stellate reticulum.
d- Enamel spindles.

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63- Which of the following appear FIRST in a developing tooth?

a- Predentin.
b- Cementoid.
c-Tome's process.
d-Secondary dentin.

64- The last hard dental tissue to be deposited is:

a- Enamel.
b-Outer layers of cementum.
c-Mantle dentin.
d- Primary dentin.

65- Which of the following is associated with an erupted tooth?

a-Outer Enamel epithelium.
b-Cervical loop.
c-Epithelial rests of Malassez.
d- Stratum intermedium.

66- Which of the following is not true of Hertwig's epithelial root sheath?

a- It lacks a stellate reticulum.
b- It is derived from stomodeal ectoderm.
c- Its remnants form epithelial rests.
d- It remains intact until a layer of cementum has been formed.

67- Which of the following contain blood vessels?

a- Cartilage.
b- Enamel.
c- Dental lamina.
d- Dental sac.

68- Which group of structures is entirely derived from the dental sac?

a- Dentin, pulp, cementum.
b- Cementum, alveolar bone, pulp.
c- Cementum, enamel, pulp.
d- Periodontal ligament, cementum, alveolar bone.

69-The development of the root begins when

a) The inner and outer dental epithelium reaching the future C.E.J.
b) The crown is completely formed.
c) The development of the supporting bone begins.
d) The development of the periodontal ligament begins.

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70- The function of the inner dental epithelium is……..

a) Exerts organization influence on the dental papillacells.
b) Lays down dentin matrix
c) Protects the enamel after eruption.
d) Provides the enamel organs with alkaline phosphatase

71-The tooth germ is composed of

a) Dental papilla, dental follicle, dental organ.
b) Dental follicle, dental lamina, vestibular lamina.
c) Dental papilla, dental lamina, dental organ.
d) Dental organ, dental follicle, dental lamina.

72-The apposition stage of tooth development begins

a) When the root of the tooth begins to develop.
b) When the first layer of dentine is deposited.
c) When odontoblast start to differentiate.
d) When the first layer of enamel lay down.

73- What statement about epithelial diaphragm is Not true?

a- It produces narrowing of the wide cervical opening.
b- Its proliferation is accompanied by proliferation of pulp cells adjacent to diaphragm.
c- It's growing end is located coronal to the root sheath.
d- It grows in a horizontal plane.

74- The cells of the stratum intermedium:

a- Have a stellate outline.
b- Are not yet differentiated in the bell stage.
c- Are in direct contact with the outer dental epithelium.
d- Are characterized by high alkaline phosphatase enzyme activity.

75-The dental follicle:

a- Is an epithelial structure.
b- Give rise to the radicular pulp.
c- Is not present in permanent teeth.
d- give rise to cementum.

76-If the process of morphogenesis is disturbed:

a- Changes will occur in the shape of crown and root.
b- Enamel formation will be abnormal.
c- Dentin formation will be abnormal.
d- Absence of cementum.

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77- The mesenchymal cells lying adjacent to the inner enamel epithelium:

a- Differentiate into ameloblasts.
b- Differentiate into odontoblasts.
c- Are under the influence of stellate reticulum.
d- Become markedly folded._

78- The dental lamina:

a- Breaks at thecap stage.
b- Has no branching.
c- Has no remnants.
d- Develops from the oral epithelium.

79- Advanced bell stage of tooth development refers to:

a- Histodifferentiation , morphodifferentiation.
b- Initiation & Proliferation.
c- Mesenchymal condensation.
d- Apposition of dental tissues.

80- The epithelial component of the tooth germ is known as:

a- Dental lamina.
b- Dental papilla.
c- Enamel organ.
d- Dental follicle.

81- Which of the following is not part of the tooth germ?

a- Dental papilla.
b- Dental follicle.
c- Dental lamina.
d- Enamel organ.

82- The dental sac plays an important role in the formation of all of the following except:

a- Periodontalligament.
b- Dentin of the root.
c- Alveolar bone proper.
d- Cementum.

83- After the crown of the tooth is formed, the dental papilla is called:

a- Dental pulp.
b- Pulp stone.
c- Dental sac.
d- Dentin.

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84- In a developing tooth, if Hertwig's sheath and the epithelial diaphragm were failed to form, the resulting tooth would exhibit:

a- Noradicular dentin.
b- Malformed or fused roots.
c- Acellular, but no cellular, cementum.
d- Defective enamel in the cervical region.

85- The reduced enamel epithelium:

a- Lacks cells derived from the stellate reticulum.
b- Contains protective ameloblasts.
c- Overlies immature enamel which has not yet reached its full thickness.
d- Does not cover enamel located at the cervical end of the crown.

86- Which of the following is a component of tooth germ?

a- Lateral dental lamina.
b- Successional lamina.
c- Vestibular lamina.
d- Dental sac.

87- The origin of enamel is:

a- Endoderm.
b- Mesoderm.
c- Ectoderm.
d- Ectomesenchyme.

88-Which statement about early dental development is incorrect?

a- Cap stage is accomplished by unequal growth in different parts.
b- In the cap stage the inner and outer enamel epithelial cells are completely alike.
c- The transition from cap stage to bell stage is particularly marked by the full differentiation of stellate reticulum.
d- The dental papilla and the dental sac develop in the cap stage.

89- The dental sac gives rise to:

a- Cementoblasts and periodontal ligament.
b- Periodontal ligament and dental pulp.
c- Dental pulp and attachment epithelium.
d- Odontoblasts and Cementoblasts.

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Answers of MCQ

Toothdevelopment

1	c	31	d	61	b
2	d	32	d	62	d
3	c	33	d	63	a
4	a	34	a	64	b
5	d	35	a	65	c
6	c	36	d	66	d
7	d	37	a	67	d
8	c	38	a	68	d
9	d	39	a	69	b
10	d	40	a	70	a
11	d	41	b	71	a
12	b	42	b	72	b
13	d	43	c	73	c
14	a	44	a	74	d
15	d	45	a	75	d
16	a	46	a	76	a
17	c	47	b	77	b
18	d	48	a	78	d
19	b	49	a	79	d
20	d	50	b	80	c
21	b	51	a	81	c
22	c	52	a	82	b
23	d	53	b	83	a
24	a	54	a	84	a
25	d	55	c	85	b
26	a	56	d	86	d
27	a	57	c	87	c
28	a	58	b	88	b
29	b	59	d	89	a
30	d	60	a		
			c		

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Embryology

1-Which of the following isnotacorrectpairingofembryonic structure and adult derivative?

a- Maxillaryprocesses: secondarypalate.
b- Primary palate: upper incisor teeth. Second
c- pharyngeal arch: mandible. Lateral lingual
d- swellings: anterior 2/3 of tongue.

2-Which of the following are all derived from the stomodeal ectoderm?

a-Salivary glands, anterior pituitary gland andenamelorgans.
b-Enamel dentin and major salivary glands.
c-Cementum, anterior pituitary gland and stratum intermedium.
d-Dental lamina, dental papilla and dental sac.

3- Which of the following does contribute to the formation of the upper lip?

a-Lateral nasal process.
b-Two maxillary processes & medial nasal process.
c-Medial nasal process.
d- First branchial arch.

4- The palatal shelves (palatine processes):

a-Are covered by endoderm.
b-Are derived from the primary palate.
c-Must assume a vertical position for palate closure to occur.
d- Are derived from the first branchial arch.

5- An oblique facial cleft is formed:

a-Due to the rupture of the buccopharyngeal membrane.
b-When the maxillary process fails to fuse with the mandibular processes.
c-When the maxillary process fails to fuse with the lateral & medial nasal process.
d-When the medial nasal process fails to fuse with the maxillary process.

6- The extrinsic muscles of the tongue:

a-Develop from the maxillaryprocess.
b-Develop from occipital myotomes.
c-Develop from tuberculum impar.
d- Develop from the second branchial arch.

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7- Which of the following embryonic structures contribute to the formation of lower lip:

a-Maxillary process.
b-Maxillary & medial nasal processes.
c-Medial & lateral nasal processes.
d- Two mandibular processes.

8- The tongue is <u>not</u> derived from:

a- Two lingual swellings.
b- Second branchial arch.
c- Third branchial arch.
d- Copula of His.

9- Palatine processes are derived from:

a- Maxillary processes.
b- Mandibular processes.
c- Copula of His.
d- Medial nasal process.

10- The maxillary processes:

a- Are covered with endoderm.
b- Are derived from the 1st branchial arch.
c- Form the primary palate.
d- Form the entire upper lip.

11- A unilateral cleft lip is formed when:

a- The palatal shelves remain in a vertical position.
b- The bucco-nasal membrane ruptures.
c- One maxillary process fails to fuse with the medial nasal process.
d- One lateral nasal process fails to fuse with the maxillary process.

12- The soft palate develops from:

a- Primary palate.
b- Palatal processes fused with the nasal septum.
c- Medial nasal process.
d- Palatal processes which are not fused with nasal septum.

13-Branchial archs sharing in tongue development are:

a) First & second.
b) First.
c) Second & third.
d) First, third & fourth.

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14- The Stomodeum is lined by

a) Endoderm. b) Mesoderm. c) Ectoderm. d) Endoderm & Mesoderm

15- The palatal shelves:

a- Develop from the lateral nasal processes.
b- From the primary palate.
c- Develop from the medial nasal processes.
d- Develop from the maxillary processes.

16- The ossification center of the developing mandible arise between:

a- inferior dental and lingual nerves.
b- mental and incisive nerves
c- mandibular and maxillary nerves.
d- superior alveolar and infra-orbital nerves.

17-The extrinsic muscles of the tongue are supplied by:

a-trigeminal nerve.
b-hypoglossal nerve.
c-facial nerve.
d- lingual nerve.

18- Secondary palate is developed from the :

a- 1st branchial arch .
b-2nd branchial arch.
c- 3rd branchial arch.
d- fronto-nasal process.

19-The muscles of mastication are derived from:

a- occipital myotomes.
b- 3rd branchial arch.
c- 1st branchial arch.
d- 2nd branchial arch.

20- palatine processes are derived from:

a- fronto- nasal process.
b-lateral nasal process.
c- 3rd branchial arch.
d- maxillary processes.

21- Tongue begins to develop at:

a-6th w.i.u. b- 11th w.i.u. c-4th w.i.u. d- 8th w.i.u.

22- Ossification center of maxilla arises between:

a-maxillary and mandibular nerves.
b-anterior superior alveolar nerve and infraorbital nerve.
c- posterior and middle superior alveolar nerves.
d- inferior alveolar and lingual nerves.

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23- First branchial arch is supplied by:
a- facialnerve.
b- glossopharyngeal nerve.
c- hypoglossal nerve.
d- mandibular nerve.
24-The neural crest cells give rise to:
a- Bone, dentine and cartilage.
b-Enamel, dentine and bone.
c- Sensory cells, enamel and dentine.
d- Fibroblasts, pericytes and enamel.

25- Which of the following are derived from Meckel's cartilage:
a- Mental ossicles, incus and malleous ofthemiddleear.
b- Lingula of the mandible and middle ear.
c- stylo-mandibular ligament.
d- All of the above.
26-The inferior parathyroid gland originates from :

a- First pharyngeal pouch.
b-First pharyngeal cleft.
c- Third pharyngeal pouch.
d- Fourth pharyngeal pouch.

27- The development of secondary palate begins from:
a- 3-4 W.I.U.
b- 4 -5 W.I.U.
c- 15-16 W.I.U.
d- 6-9 W.I.U.

28- The occipital somites myotomes give rise to
a- Muscles of mastication b- Muscles of the lip.
c- Muscles of the tongue.
d- Muscles of cheek.

29- The malleus of inner ear originates from:
a- Condylar cartilage. b- Coronoid cartilage.
c- Meckel's cartilage. d-None of the above.

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Answers of MCQ

EMBERYOLOGY

1	c	15	d
2	a	16	b
3	b	17	b
4	d	18	a
5	c	19	c
6	b	20	d
7	d	21	c
8	b	22	b
9	a	23	d
10	b	24	a
11	c	25	a
12	d	26	c
13	d	27	d
14	c	28	c
		29	c

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Eruption

1- Which of the following pairsofmovementspredominate during the post-eruptive period?

a-Distal and lingual.
b-Axial and mesial.
c-Distal and axial.
d-Axial and rotational.

2- Passive eruption is:

a-Growth in the length of the root.
b-Apical migration of the junctional epithelium.
c-Deposition of bone at the fundus.
d-The formation of cellular cementum.

3- Which of the following does <u>not</u> occur during the eruptive phase?

a-Dentin deposition.
b-Narrowing of apical foramen.
c-Increased thickness of cementum.
d- Enamel formation.

4- During the pre-eruptive phase of tooth eruption:

a- Cementoblasts begin to secrete cementoid.
b- The apical foramen is beginning to close.
c- The developed tooth at the bell stage lies within a crypt of bone.
d- The root is formed.

5- Which of the following occurs during the pre-eruptive stage of eruption of a tooth:

a- Differentiation of the stratumintermedium.
b- Formation of the dental sac.
c- Formation of stellate reticulum.
d- Maturation of enamel.

6- Active eruption of the tooth:

a- Begins as soon as the roots have begun to develop.
b- Begins when the apical cementum is deposited on the roots of the tooth.
c- Begins as the occlusal tip starts moving towards the occlusal plane.
d- Continues until exposure of the cementum has occurred.

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7- The root formation theory of eruption proposes, that eruption is due to the push of the:

a- Expanding pulp intheroot canals againstthealveolarcrypt.
b- Elongating roots against the base of the alveolar crypt.
c- Elongating roots against the Hammock ligament.
d- Proliferating periodontal ligament.

8-Actual eruptive movements of tooth occur mainly in a :

a) Horizontal direction.
b)Complex pattern.
c) Axial direction.
d) Rotational direction.

9-Which of the following does not occur in the active phase of tooth eruption:

a-Organizationofaperiodontalligamentfrom thedental follicle.
b- Gradual separation of attachment epithelium from the enamel surface.
c- Root formation.
d- Occlusal wear.

10-The most likely factor responsible for tooth eruption is:

a- Vascular pressure.
b- The developing periodontal ligament.
c- Bone growth.
d- The growing root.

11-All of the following statements about the eruption of a permanent tooth and the exfoliation of its predecessor are true, except:

a- Priorto eruption, the permanent tooth is completelyformed.
b- Resorption of deciduous roots is an intermittent process.
c- Earliest resorption of deciduous roots is on root surface, facing the permanent tooth.
d- Eruptive movements of the permanent tooth may not be solely responsible for deciduous root resorption.

12-When a tooth first erupts into the oral cavity, the attachment epithelial cuff is composed of epithelium derived from:

a- Dentallamina.
b- Epithelial rests of Malassez.
c- Reduced dental epithelium.
d- Epithelial root sheath of Hertwig's.

13- In the posteruptive phase a tooth undergoes primarily:

a- Rotational movement.
b- Movements in a distal direction.
c- Movements in an apical direction.
d- Movements in an axial & mesial direction.

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14- During tooth eruption, the permanent successors of the primary teeth move occlusaly and:

a- Mesially.
b- Buccally.
c- Lingually.
d- Distally.

15- During tooth eruption into the oral cavity, what happens to the epithelial covering of the enamel:

a- It remainsas the primarycuticle.
b- It units with the oral epithelium and then degenerates. c- It is replaced by oral epithelium and then degenerates. d- It units with the oral epithelium and becomes primary cuticle.

16- During tooth eruption, bone formation is often seen at the base of socket. This newly formed bone is usually in the form of:

a- Coarsefiberedwovenbone.
b- Compact bone.
c- Radiating bone.
d- Horizontal trabeculae.

17-The pre-eruptive phase:

a- Begins in the late bell stage.
b- Begins in the early bell stage.
c- Begins in the beginning of root formation.
d- Begins in the cap stage.

18- Types of movement during the pre-eruptive phase:

a- Axial- eccentric.
b- Eccentric-tilting.
c- Bodily-eccentric.
d- Bodily- axial.

19-Eruptive phase:

a- Accompanied by bleeding of the gingival.
b- No discomfort or pain.
c- Eruptive teeth move at different speeds, at different times.
d- Has only axial occlusal movement.

20-Histological changes occurring during the eruptive phase:

a- Root formation only.
b- Formation of the attachment apparatus only.
c- Dento- gingival junction appearance only.
d- All of the above.

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Answers of MCQ

ERUPTION

1	b	11	a
2	b	12	c
3	d	13	d
4	c	14	b
5	d	15	b
6	a	16	d
7	d	17	b
8	c	18	c
9	d	19	c
10	b	20	c

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Shedding

1- During shedding which of thefollowingiswrong:

a- The pulp plays an active role.
b- The pulp plays a passive role.
c- The pulp appears histologically normal till the end of the process.
d- The neural elements remain in the pulp until shedding is completed.

2- Which of the following is true about ankylosis?

a- Is a union of the tooth with the periodontal ligament.
b- The active eruption of an ankylosed tooth continues.
c- Is a bony union between the alveolar bone and the roots of teeth.
d- Ankylosed teeth do not show Howship's lacunae on the root.

3- Shedding is predominantly the:

a- Physiologic loss of impacted teeth.
b- Pathologic loss of non-vital primary teeth.
c- Physiologic loss of permanent teeth.
d- Physiologic loss of primary teeth.

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Answers of MCQ

Shedding

1	b
2	c
3	d

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Enamel

1- Ameloblasts having ruffled bordersare:

a-Secretory ameloblasts.
b-Maturative ameloblasts.
c-Protective ameloblasts.
d-Transitional ameloblasts.

2-Which of the following is <u>not</u> a feature of enamel?

a- Contourlines ofOwen.
b- Cross striations.
c- Prism sheath.
d- Perikymata.

3-Maturative ameloblasts are characterized by having:

a- A distally positioned nucleus.
b- A highly basophilic cytoplasm.
c- A highly-developed Golgi complex.
d-A distal cell membrane with many infoldings.

4- Which of the following does <u>not</u> occur during maturation of enamel?

a- Influx of calciumand phosphorus.
b- Growth of hydroxyapatite crystals.
c- Increase in organic content.
d-Decrease in the water content.

5- Incremental lines are a result of variation in:

a- Proliferation.
b- Histodifferentiation.
c- Morphodifferentiation.
d- Rhythmic apposition.

6- The prismatic structure of enamel is due to:

a-Variation in thedensity of organicmaterial.
b-Variation in crystal orientation.
c-A rhythmic pattern of apposition.
d- The periodic change in prism direction.

7- The prismless layer of enamel:

a-Is found only in primary teeth.
b-Does not contain prism & prism sheaths.
c-Is found near the amelo-dentinal junction.
d-Contains no hydroxyapatite.

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8- Enamel tufts:

a- Contain cellular processes.
b- Can pass to dentin.
c- Are seen in decalcified sections.
d- Are grass like structure.

9- During the maturation stage of enamel formation, which of the following does <u>not</u> occur in the ameloblastic layer?

a-Water absorption.
b-Protein absorption.
c-Change in cellular organelles.
d-Secretion of enamel matrix.

10- When prismless enamel is formed, the secretory ameloblast does <u>not</u> contain:

a-Rough endoplasmic reticulum.
b-Tome's process.
c- Nucleus.
d- Secretory granules.

11- Which of the following is <u>true</u>:

a-Enamel is the weakly mineralized tissue.
b-Enamel is the highly mineralized tissue. _
c-Enamel is the highly permeable tissue.
d-Enamel is a non brittle tissue.

12- Neonatal lines are present in:

a-All permanent teeth.
b-Permanent canines.
c-All deciduous & first permanent molar.
d- Premolars.

13- Organic components of enamel are primarily removed from enamel by:

a-Secretory ameloblasts.
b-Maturative ameloblasts.
c-Transitional ameloblasts.
d-Enamel spindle.

14- The main difference between the rod and the interrod region is:

a-Crystal size.
b-Crystal orientation.
c-Crystal composition.
d-In the nature of the organic component.

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15- The striae of Retzius:

a- Are parallel to the enamel rods.
b- Are parallel to the surface of the enamel in the cervical region.
c- Are not seen in the cuspal one third of the enamel.
d- Would be seen as concentric rings in a horizontal section of the crown.

16-Enamel is composed of:

a) Rod, interrod & rod sheath.
b) Only enamel prisms.
c) Needle shaped crystals.
d) Prism sheath.

17-The extensions from secretory ameloblast cells that give structure to enamel rods are called:

a) Enameltuft.
b) Tomesprocesses.
c) Enamelspindles.
d) Enamellamellae.

18- Striae of Retzius:

a) Resultfromenamelcrack.
b) Separateprenatalfrompostnatalenamel.
c) Areduetorhythmicappositionofenamel.
d) Arehypermineralizedstructure.

19- Enamel Rods:

a) Theirnumbervariesindifferentteeth.
b) Haveanaveragewidthof1um.
c) Are Presentattheenamelsurface.
d) Haveastraightcourseatthecuspsofteeth.

20- Enamel is formed of:

a- 76%minerals&24%organicmaterialsandwater.
b- 96%minerals&4%organicmaterialsandwater.
c- 86%minerals&14%organicmaterialsandwater.
d- 99%minerals&1%organicmaterialsandwater.

21- which of the following is absent in all premolars:

a- Enamelspindle.
b- Enameltuft.
c- Neonatalline.
d- Enamellamella.

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22- Cross striations in enamel:

a- Arean optical phenomena.
b- Result from abrupt change in environment before and after birth.
c- Result from mineralization disorder.
d- Appear to be periodic bands at 4 microns interval across the rod.

23- The prism sheath:

a- Is adense continuous sheath separating rod from inter-rod.
b- Result from abrupt change in crystals orientation.
c- Is more mineralized than rod and inter-rod.
d- Found in prismless enamel.

24- The complex arrangement of rods over the cusps of teeth is termed:

a- Prismless enamel.
b- Gnarled enamel.
c- Prenatal enamel.
d- Postnatal enamel.

25- The origin of enamel is:

a- neuralcrest cells
b- Ectoderm.
c- Mesoderm.
d- Ectomesenchyme.

26- Hypocalcified areas in enamel are:

a- Enamel spindles.
b- Dentino-enamel junction.
c- Interprismatic substances.
d- Enamel tufts.

27- Prismless enamel:

a- Is less mineralized than prismatic enamel.
b- Is found only in the cervical area of permanent teeth.
c- Is harder than prismatic enamel.
d- Found near the dentino-enamel junction.

28- The perikymata:

a- Is an optical phenomena resulting from changes in direction of enamel rods.
b- Is a highly mineralized outer enamel surface.
c- Is formed of transverse grooves, represents an outer manifestation of brown stria of Retzius.
d- Found at the cusp tip.

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29- What is the result of cracks before tooth eruption:

a- Enamel lamella type A.
b- Enamel lamella type B.
c- Enamel lamella type C.
d- Enamel tufts.

30- The primary enamel cuticle is formed by the:

a- Outer dental epithelium.
b- Ameloblasts just before tooth eruption.
c- Fusion of oral epithelium and reduced enamel epithelium.
d- Ameloblasts after the enamel has been completed.

31- The surface enamel:

a- Is similar to thesubsurface enamel.
b- Is softer than subsurface enamel.
c- Contains less mineral than subsurface enamel.
d- Highly mineralized than subssurface enamel.

32- The prism sheath:

a- Is a continuous sheath.
b- Is more calcified than the enamel rod.
c- Is less calcified than the enamel rod.
d- Is equally calcified when compared to the enamel rod.

33- The predominant inorganic portion of enamel is:

a- Calcium.
b- Carbonate.
c- Phosphate.
d- Fluoride.

34- Incremental lines of Retzius form grooves on the surface of the enamel termed:

a- Perikymata.
b- Neonatal lines.
c- Gnarled enamel.
d- Enamel cuticle

35- Which of the following is poorly calcified enamel?

a- Gnarled enamel.
b- True enamel lamellae.
c- Perikymata and gnarled enamel.
d- Interprismatic substance.

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36- The cross striations characteristic of enamel rods represent:

a- Incremental growth pattern.
b- The junctional complex of ameloblasts.
c- Hunter-Schreger bands.
d- The inter rod region.

37- Enamel spindles:

a- Are areas of prismatic enamel.
b- Are present at the occlusal surfaces of enamel.
c- Are the terminal ends of odontoblastic processes.
d- Are located at the cemento-enamel junction.

38- Which of the following is a <u>true</u> statement about the organic matrix of the newly synthesized enamel?

a- Itrepresents 30%by weightoftheorganiccontent.
b- Ithas a high content of amelogenins.
c- Itcontains collagen fibers.
d- Ithas a high inorganic content.

39- Which of the following is the source of nutrient supply to the secretory ameloblasts during early crown formation?

a- Subodontoblasticplexus of capillaries.
b- Central blood vessels in the dental papilla.
c- Dental sac capillaries adjacent to the epithelial diaphragm.
d- Dental sac capillaries adjacent to the outer dental epithelium.

40- Tome's processes:

a- Contain mitochondria, but lack Golgi saccules and rough endoplasmic reticulum.
b- Lack secretory granules.
c- Indent the immature enamel to produce a serrated outline on its appositional surface.
d- Are located at the proximal end of the ameloblast.

41- Enamel spindles are formed by:

a- Odontoblastic process.
b- Ameloblastic process.
c- Hypocalcified rods.
d- Cracks.

42- Neonatal lines are found in all the following <u>except</u>:

a- Enamel of primary incisors.
b- Enamel of permanent first molar.
c- Enamel of permanent second molars.
d- Enamel of primary molars.

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43- Hypocalcified ectodermal structures in enamel are:

a- Spindles and lamellae.
b- Odontoblastic processes and tufts.
c- Tufts and spindles.
d- Lamellae type A,and tufts.

44- Which of the following most correctly describes the significance of Hunter Schreger bands?

a- It indicates variation in size of enamel rods.
b- It reflects variations in the length of enamel rods.
c- It represents hypercalcified areas in cervical enamel.
d- It is an optical effect that is not related to variations in enamel structure.

45- The orientation of hydroxyapatite crystals in enamel, the greatest crystal density would be present in:

a- enamel lamellae.
b- The interrod region.
c- The prismless layer.
d- Gnarled enamel.

46- Perikymata:

a- Are oriented parallel to the long axis of the crown (tooth).
b- Are surface manifestation of Retzius lines.
c- Are only located in prismless enamel.
d- Produce the scalloping seen along the DEJ.

47- Unmineralized structures seen in the enamel are :

a- Cross striations.
b- Enamel tufts.
c- Incremental lines of Retzius.
d- Enamel spindles.

48- All of the following statements about the incremental lines of Retzius in enamel are true <u>except</u>:

a- They run as concentric lines overthe cusp.
b- Perikymata are outer manifestations of it.
c- They are formed as result of rhythmic apposition.
d- They do not reach the cervical region.

49-Incremental lines of Retzius appear:

a- In dentin.
b- In enamel.
c- In cementum.
d- In bone.

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50-Primary enamel cuticle is:

a- Afibrillar cementum.
b- Precipitate of salivary proteins.
c- Relatively structureless layer of enamel.
d- A last product of ameloblasts.

51- Hunter-Schreger bands have all the following characteristics <u>except</u>:

a- They are best seen in reflected light.
b- They are alternating dark and light bands
c- They are most likely an optical phenomenon
d- They are found at the cusp tip.

52- Enamel tufts and enamel lamellae resemble each other in that both:

a- Are formed prior to tooth eruption only.
b- Extend over full thickness of the enamel.
c- Are hypermineralized structures.
d- Are hypomineralized structures._

53-Enamel is by weight:

a- 70% mineral, 26% organic matrix, 4% water.
b- 76% mineral, 20% organic matrix, 4% water.
c- 25% mineral, 12% organic matrix, 3% water.
d- 95% mineral, 1% organic matrix, 4% water.

54- All of the following structures in enamel are ectodermal origin except:

a- Enamel spindle.
b- Enamel prisms.
c- Enamel lamellae (type A).
d- Enamel tufts.

55- Enamel is:

a. An ectodermal tissue covering the anatomical crown.
b. A mesodermal tissue covering the crown.
c. An endodermal tissue.
d. A mesenchymal tissue.

56- Gnarled enamel has the following criteria <u>except</u>:

a. Found under cusp tips and incisal ridges.
b. The course of the enamel rods is more straighter.
c. The enamel rods become twisted and braided.
d. Found to give maximum strength to the areas subjected more to the masticatory forces.

57- The peripheral part of the E rod is called:

a. Interrod region. b. Rod sheath.
c. Rod region. d. Key hole.

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58- Concerning the rod sheath, one of the following statement is incorrect:

a.It occupies theperipheral part of theE.rod.
b.Itformstheborderwherecrystalsoftherodand interrod enamel meet.
c.It ishighlycalcifiedthantherod.
d.Ithasahigherorganiccontentthusresistsdecalcification more than the rod.

59- Theinterrodenamelis:

a.More calcifiedthantheE.rod.
b.Less calcifiedthantheE.rod.
c.The same calcificationas the E.rod.
d.Organicinnature.

60- Crossstriationsorshortincrements:

a.Representtherhythmicappositionofthe E.matrix by ameloblasts.
b.Representalternatingconstrictionandexpansion of E. rods at and in-between the striations.
c.Representthedailyrateofsecretoryactivityof the ameloblast.
d.All of theabove.

61- The formation of the enamel matrix between two short increments needs:

a.One day. b.Twodays.
c.Threedrays. d-Four days.

62- Neonatallineisallthefollowingexcept:

a.Is oneof thebrownstriaeof Retzius that gets accentuated.
b.Ispresentindeciduousteethandallpermanent molars.
c.Separatesenamelformedbeforebirthfromthat formed after birth.
d.Itisduetoabruptchangeinnutritionandenvironmental condition at birth.

63- ThequalityoftheprenatalEnamel:

a.Is better thanthat of thepostnatal E.
b.Isnot as better asthat of thepostnatal E.
c.Is equal tothequalityof postnatal E.
d.Not relatedtothequalityof postnatal E.

64- Thedentinoenameljunction(DEJ):

a.Appearstobescallopedwithconvexitiesofthe scallops directed to E.
b.Appearstobescallopedwithconvexitiesofthe scallops directed to D.
c.Appearstobescallopedwithconvexitiesofthe scallops directed to cementum.
d.Non of theabove.

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65- The firm attachment between E and D is due to:

a.Under cuts created by the scallops of the DEJ.
b.Interdigitation at the DEJ between fibrils of first formed layer of D and that of first formed layer of E.
c.Interdigitation between the hydroxyapatite crystals that are deposited during mineralization of E and D.
d.All of the above.

66- E. lamellae are all the following <u>except</u> :

a.Thin leaf or sheet like structures.
b.Extend from the E surface and may extend to D.
c.Demonstrated in the longitudinal ground section only.
d.Hypo or unmineralized structures

67- Enamel spindles:

a.They are odontoblastic cell bodies pass across the DEJ into the E.
b.They are extended between the preameloblasts after enamel is formed.
c.They are most common beneath cusps.
d.They are best demonstrated in longitudinal decalcified section.

68- Rodless Enamel:

a.Occurs in the outermost 70 microns of the E.
b.Their crystals are aligned at right angel to the surface
c.Found most common at the cervical areas of E surface.
d.All of the above.

69- Perikymata are:

a.Vertically oriented wave like grooves.
b.The external manifestation of the brown striae of Retzius.
c.Numerous at the incisal or cusp regions.
d.Parallel to the AEJ

70- Which of the following tissues begins formation <u>first</u>:

a.Enamel. b. Dentin.
c.Pulp. d. Cementum.

71- Attrition is characterized by:

a.Being a condition of normal wear of the occlusal and proximal surfaces.
b.A chemical loss of occlusal tooth structure, but not from dental caries.
c.A loss of crown structure on the occlusal surface by mechanical means.
d.None of the above.

72- In mature dental tissues, the only tissue whose cells are lost is:

a. Dentin. b. Enamel.
c. Cementum. d. Pulp.

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73- Incremental lines of Retzius appear:

a. In dentin and follow the appositional pattern.
b. In enamel and follow the appositional pattern.
c. In cementum and follow the appositional pattern.
d. As lines in E running at right angles to the E surface.

74- Enamel is by weight:

a. 88-90 % mineral, 10-12 % organic matrix and water.
b. 76% mineral, 20% organic matrix, 4 water.
c. 25% mineral, 12% organic matrix, 3% water.
d. 96% mineral, 1-2% organic material, 2-3% water.

75- Which of the following is <u>not</u> true about Enamel rod ?

a. They run from the dentinoenamel junction to the crown surface.
b. Length of all enamel rods is equal.
c. They are the basic structural units of E.
d. They may be hexagonal ,rounded, oval or keyhole shape in cross section .

76- Which of the following statements about E . is <u>false</u>?

a. The heads of enamel rods are directed cervically, the tails occlusally.
b. Rod sheaths are relatively hypocalcified areas surrounding the rods.
c. Human E rods in cross section have an outline similar to a keyhole.
d. Hydroxyapatite crystals run parallel to the long axis of the rods and perpendicular to it in the interrod region .

77- It is particularly difficult to finish the wall of a cavity preparation in cuspal enamel because of the presence in that area of:

a. Tufts. b.HorizontalE.rods.
c. Gnarled E. d. Spindles.

78- Enamel tufts and Enamel lamella resemble each other, in that both:

a. Are formed after to tooth eruption.
b. Extend over full thickness of the enamel.
c. Are hypermineralized structures.
d. Are oriented in a vertical direction perpendicular on the DEJ.

79- All of the following statements about the incremental lines of Retzius in Enamel are true <u>except</u>:

a. Inlongitudinal section they run obliquely, relative to D.E.J.
b. Perikymata in newly erupted teeth are manifestations of the lines of Retzius.
c. They are formed as a result of rhythmic variations in the calcification.
d. Under no conditions can a line of Retzius to be a neonatal line.

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80- E. has no possibility of self repair because:

a.Ithasasmall percentage of organic content.
b.Itsformative cells are lost once it is fully formed and the tooth erupted.
c.Itisessentially a keratin tissue and has no blood vessels.
d.Ithasnodirect connection with the active cells of the pulp tissue.

81- At the time Enamel matrix is first formed in a tooth:

a.Nopredentinhas yet been formed.
b.Rootofthe tooth is largely covered with acellular cementum.
c.Nucleiofameloblasts move to non secreting end of the cell.
d.Cellsofstellate reticulum near the cervical portion of the tooth have lost their nuclei.

82- Hypomineralized structures which extend from the D.E.J to the surface of the Enamel are called:

a. Hunter schregerbands.
b. E. spindle.
c. Incremental lines of Retzius.
d. Enamel lamellae.

83- Maturation of Enamel is characterized by a percentage increase in inorganic content and a percentage:

a. Increase in water.
b. Decrease in water and organic content.
c. Increase in water and decrease in organic content.
d. Decrease in water and increase in organic content.

84- The stimulus that initiates the actual formation of Enamel matrix seems to be the:

a. Presence of predentin.
b. Reduction of stellate reticulum.
c. Influence of increased vascularization of the dental sac.
d. Reversal of functional polarity of ameloblast.

85- In a routine demineralized histological section of an adult tooth, the area representing Enamel is generally optically clear because:

a. The mounting mediumhas the same refractive indexasenamel.
b. E is very brittle and thus breaks off during sectioning.
c. Ordinary stains don't stain the organic matrix of E.
d. Enamel is highly calcified.

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86- A layer of cells that seems to be essential to enamel formation but does not actually secrete the enamel is:

a. Innerdental epithelium.
b. Stratum intermedium.
c. Reduced dental epithelium.
d. Vestibular lamina.

87- Which of the following are poorly calcified Enamel:

a. Gnarled E .and E rods.
b. E. lamellae and incremental lines.
c. Perikymata and gnarled E.
d. E. rods and incremental lines.

88- Enamel matrix is laid down by ameloblasts:

a. Finally contains about 80% by weight inorganic matter.
b. Includes organic matter that is primarily collagen. c. Persists as a vital tissue through the life of a tooth. d. Non of the above.

89- Enamel tufts:

a. Begin on the outer edge of E .
b. Are developmental defects and are worn away by masticatory forces.
c. Have a brush like appearance and are hypo calcified areas in E.
d. Represent hypercalcified E. rods.

90- The rod sheath:

a. Has cross striations.
b. Is more calcified than the E. rod.
c. Is less calcified than the E. rod.
d. Has lesser amount of organic matter than the E. rod.

91- Ionic exchanges between erupted Enamel .and Saliva:

a. Don't occur as the formative cells of the E .are lost once it is formed.
b. Occur only for two to five years following eruption.
c. Cease at puberty.
d. Occur throughout life.

92- The predominant organic portion of mature Enamel. is:

a. Calcium. b. Fluoride.
c.Enamelin. d. Phosphate.

93- Fluorine deposition in Enamel:

a. Increase with age.
b. Decrease with age.
c. Sometimes increases or decreases with age.
d. Not affected by age.

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94- Which of the following organic components are present in Enamel:

a. Carbohydrates. b. Fat.
c. Proteins. d. All of the above.

95- The rodless layer of enamel:

a-Is found only in primary teeth.
b-Does not contain rod and rod sheaths.
c-Is found near the amelo-dentinal junction.
d-Contains no hydroxyapatite.

96- Hunter-Schreger bands are:

a-found in the outer surface of enamel.
b-run parallel to the striae of Retzius.
c-best visualized in transverse section of the crown.
d-reflect periodic changes in prism direction of enamel.

97- The direction of enamel rods in general is:

a-parallel to the outer surface of enamel.
b-parallel to dentine.
c-perpindicular to the amelo-dentinal junction.
d-random.

98- Which of the following contains the highest concentration of hydroxyapatite:

a-cementum.
b-enamel.
c-primary dentine.
d-secondary dentine.

99- Enamel prism of the cusp tip passes:

a- in wavy course.
b- in s-shape course.
c- in straight course.
d- in twisted course.

100-Developmental enamel lamellae is:

a- type A enamel lamellae.
b- type B enamel lamellae.
c- type C enamel lamellae.
d- accidental enamel crack.

101- The structures which are <u>not</u> seen in a decalcified section are:

a-Dentinal tubules.
b-Incremental lines of Retzius.
c-Secondary dentin.
d-Cellular cementum.

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Answers of MCQ

Enamel

1	b	31	d
2	a	32	c
3	d	33	a
4	c	34	a
5	d	35	b
6	b	36	a
7	b	37	c
8	d	38	b
9	d	39	d
10	b	40	c
11	b	41	a
12	c	42	c
13	b	43	d
14	b	44	d
15	d	45	c
16	a	46	b
17	b	47	d
18	c	48	d
19	a	59	b
20	b	50	d
21	c	51	d
22	d	52	d
23	b	53	d
24	b	54	a
25	b	55	a
26	d	56	b
27	c	57	b
28	c	58	c
29	b	59	c
30	d	60	d

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61	a	91	d
62	b	92	c
63	a	93	a
64	b	94	d
65	d	95	b
66	c	96	d
67	c	97	c
68	d	98	b
69	b	99	d
70	b	100	a
71	a	101	b
72	b		
73	b		
74	d		
75	b		
76	a		
77	c		
78	d		
79	d		
80	b		
81	c		
82	c		
83	b		
84	a		
85	d		
86	b		
87	b		
88	d		
89	c		
90	c		

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Dentin

1-Interglobular spaces:

a-Do not contain dentinal tubules.
b-Do not contain type I collagen.
c-Are formed during the course of globular mineralization.
d- Are seen only in secondary dentin.

2- Which would <u>not</u> be present in a ground section?

a-Inorganic crystals.
b-Odontoblast.
c-Interglobular space.
d- Dentinal tubules.

3- A major difference between secondary and primary dentine is that:

a- There is no predentine.
b- There is no hydroxyapatite.
c- There is no collagen fibers.
d- The dentinal tubules pass in wavier course through secondary dentine.

4- The interglobular dentin is found:.

a-Near the amelo-cemental junction.
b-In the root dentin.
c-Near the amelo-dentinal junction.
d-Near the dentino-cemental junction.

5- Which of the following is <u>MOST</u> likely to be found in the dentinal tubules:

a-Tomes' fibers.
b- Oxytalan fibers.
c-Tomes' process.
d- Thick collagen fibers.

6- Primary dentin differs from both secondary and reparative dentin in that:

a- Its matrix is calcified.
b- Its inorganic component is hydroxyapatite.
c- It usually has a greater number of tubules per unit area.
d- It is produced by mesenchymal cells.

7- Which of the following is <u>not</u> present within dentin?

a-Cell bodies. b-Hydroxyapatite. c-Collagen type I. d-Incremental lines.

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8- Which of the following is <u>true</u> of peritubular dentin ?It :

a-Is less mineralized thanintertubulardentin.
b-May be considered as intratubular dentin.
c-Is best visualized in decalcified sections.
d-Is found in the interglobular dentin.

9- Which of the following is <u>not</u> present in odontoblasts?

a-Golgi saccules and vesicles.
b-Rough endoplasmic reticulum.
c-Desmosome-like junctions.
d- Numerous processes forming a striated border.

10- Human primary dentin contains:

a-Mantle & circumpulpaldentin.
b-Tomes' process.
c-Cell bodies.
d-Atubular dentin.

11- Which of the following is considered as age changes?

a-Transparent dentin.
b-Cellular cementum.
c-Primary dentin.
d- Enamel spindle.

12- Which of the following is <u>true</u>:

a-Alayer of predentin is found between primary and secondary dentin.
b-Dentinal tubules do not pass through interglobular dentin.
c-The physiological secondary dentin contain dentinal tubules.
d-Many large collagenous fiber bundles are found in the dentinal tubules.

13- Reparative dentin:

a-Contain regular and numerous dentinal tubules.
b-Obliterates the apical foramen.
c-Forms on localized regions of the pulpal surface of the dentin.
d-Formation is induced by the cells of the inner dental epithelium.

14- Matrix vesicles are present during:

a-The initial mineralization of dentin.
b-The initial mineralization of enamel.
c-The initial mineralization of cementicles.
d-The initial mineralization of denticles.

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15- Which of the following is <u>least</u> prominent in odontoblasts and secretory ameloblasts?

a- Rough endoplasmicreticulum.
b- Mitochondria.
c- Golgi apparatus.
d- Lysosomes.

16- Collagenous fibers are <u>not</u> found in:

a- Mantle dentin.
b- Reparative dentin.
c- The outer prismless layer.
d- The dental sac.

17- Obliteration of the dentinal tubules by mineral deposits:

a- Occurs only when the dentin becomes exposed to the oral fluids.
b- Produces dead tracts.
c- First occurs in predentin.
d- Is seen in sclerotic and transparent dentine.

18- Tome'sgranularlayer is:

a- An extension of odontoblast processes into enamel.
b- A region lacking dentinal tubules.
c- Following the incremental lines.
d- Derived from the epithelial rests of Malassez.

19- Amelogenesis and dentinogenesis are similar in which respect?

a- Both processes involve the secretion of collagenous fibers
b- impregnation with hydroxy-apatite crystals.
c- Both processes involve maturation with hydroxyappatite crystals.
d- Both processes start simultaneously at the early crown stage.
Both processes continue throughout the life of the tooth.

20- Which of the following is <u>wrong</u>?

a- The formation of primary dentin continues throughout life.
b- Primary dentin contains a significantly higher number of tubules per unit
c- volume.
d- Reparative dentin forms as a response to injury.
Dead tracts are more common in secondary than in primary dentin.

21- Which of the following has the highest mineral content?

a- Predentin.
b-Peritubular dentin.
c-Intertubular dentin.
d- Cementum.

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22-Dentin is a variety of Connective tissue because it:

a- iselastic and contains mucopolysaccharides.
b- contains collagen fibers and cell processes derived from mesenchyme.
c- contains inorganic cells.
d- has few cells and much intercellular material.

23-Cementum is a tissue that:

a- provides attachmentof periodontal ligament fibers.
b- is easily resorbed than bone.
c- it contains blood vessels.
d- it contains neural receptors.

24-Cementum deposition cervically on enamel surface is due to:

a- premature degeneration of epithelial root sheathofHertwig.
b- loss of reduced enamel epithelium.
c- persistence of epithelial root sheath of Hertwig.
d- cementoblasts coming from dental sac.

25-Neonal line is present in:

a- enamel and cementum. b- dentin and cementum.
c- enamel and dentin. d- non of the above.

26-Obliteration of dentinal tubules by minerals will result in:

a- formation of dead tracts.
b- formation of secondary dentin.
c- formation of sclerosed dentin.
d- formation of irregular or reparative dentin.

27-Age changes will increase:

a- dentinsclerosis. b- interglobular dentin.
c- in cementum thickness. d- Tome's granular layer.

28-formation of matrix vesicles in calcification of dentin is limited to:

a- circumpulpal dentin. b- Mantle dentin.
c- all types of dentin. d- non of the above.

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Answers of MCQ

Dentine

1	c	15	d
2	b	16	c
3	d	17	d
4	c	18	b
5	a	19	b
6	c	20	a
7	a	21	b
8	b	22	b
9	d	23	a
10	a	24	b
11	a	25	c
12	c	26	c
13	c	27	a
14	a	28	b

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Cementum

1- Cementum may cover part of the enamel due to:

a-Premature differentiation of cementoblasts.
b-Destruction of the reduced enamel epithelium covering this part.
c-Malfunctioning of the maturative ameloblasts.
d- Premature destruction of the epithelial root sheath of Hertwig.

2- Which of the following cell is responsible for cementogenesis?

a- Ameloblast.
b- Osteoblast.
c- Cementoblast.
d-Odontoblast.

3- Which of the following is <u>true</u> about cementum? It:

a-Provides nutrientstosecondary dentin in theroot.
b-Provides for the attachment of principal periodontal fibers.
c-Contains neural receptors.
d-Is more easily resorbed than bone.

4- Which of the following is <u>not</u> a feature of cellular cementum?

a-Lacunae and canaliculi.
b-Continuous remodeling.
c-Sharpey's fibers.
d- Incremental lines.

5- Cementum may be deposited on the surface of enamel as a consequence of:

a-Early degeneration of Hertwig's sheath.
b-Injury to the reduced enamel epithelium.
c-Late degeneration of Hertwig's sheath.
d-During passive eruption.

6- Cementoblasts:

a-Are derived from the dental papilla.
b-Secrete both collagen and glycosaminoglycans.
c-Are multinucleated cell.
d- May become incorporated into acellular cementum.

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7- Which of the following is <u>not</u> present inside both cellular and acellular cementum:

a- Cementoblasts
b- Hydroxyapatite crystals.
c- Incremental lines.
d- Sharpey's fibers.

8- Which of the following does <u>not</u> increase normally with age?

a- Pulp stones.
b- Dentin sclerosis.
c- Interglobular dentin.
d- Cementum thickness.

9-human teeth contain varities of cementum:

a- acelllular extrinsic fiber cementum.
b- cellular intrinsic fiber cementun.
c- mixed stratified cementum.
d- all of the above.

10-Incremental lines of Salter are:

a- highly mineralized areas having less collagen and great amount of mucopolysaccharides.
b- hypomineralized areas containing less collagen and great amount of mucopolysaccharides.
c- unmineralized areas containing no collagen nor mucopolysaccharides.
d- similar to cementum mineralization.

11-cementum hypertrophy occurs:

a- in non functioning teeth.
c- in embedded teeth.
b- functioning teeth.
d- in malposed tooth

12-Cementoblasts are derived from:

a- epithelial root sheath of Hertwig.
b- epithelial rests of Malassez.
c- cells of dental papilla.
d- ectomesenchymal cells of tooth sac.

13- Persistance of epithelial root sheath of Hertwig will result in:

a- Formation of intermediate cementum.
b- No cementum formation.
c- Formation of acellular cementum.
d- Formation of accessory root canal.

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Answers of MCQ

CEMENTUM

1	b	8	c
2	c	9	d
3	b	10	a
4	b	11	b
5	b	12	d
6	b	13	b
7	a		

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Pulp

1- The nerve supply within the dentalpulpprovides for:

a- Pain perception, vascular control and proprioception.
b- Thermal perception, vasodilatation and cementum sensitivity.
c- Pain perception, dentin sensitivity and vascular control.
d- Proprioception, taste and vasoconstriction.

2- With age, the volume of the pulp decreases because of:

a-A decrease in the number of collagenous fibers.
b-The formation of secondary dentin.
c-Sclerosis of radicular dentin.
d-A decrease in number of fibroblasts.

3- Thetruepulp stone isformed usually:

a- In large size.
b-In the coronal pulp.
c- In the radicular pulp.
d-Around degenerated nidus.

4- Thepulp sensation is:

a- Touch.
b- Propioceptive.
c- Pain sensation.
d- Cold.

5- As a result of aging, the dental pulp in human permanent teeth:

a- Increase in size andbecomes more vascular.
b- Becomes less cellular and more fibrous. Becomes more cellular and contains more ground substance. Becomes more vascular and contains larger neural elements. **6- An embedded pulp stone is most likely to be surrounded by:**

a- Fibrous tissue.
b- Primary dentin.
c- Secondary dentin.
d- Odontoblasts.

7- Which of the following is thought to protect the dental pulp?

a- Sclerotic dentin.
b- Dead tracts.
c- Tome's granular layer.
d- Denticles.

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8- A false denticle:

a- Consists of dentinal tubules.
b- Is usually associated with a dead tract.
c- Indicates a dead or dying pulp organ.
d- Consists of concentric layers of calcification.

9- Dental pulp develops from:

a- Dental organ.
b- Dental sac.
c- Dental papilla.
d- Dental lamina.

10- Nucleus with cart wheel appearance is found in:

a- Plasma cells.
b- Lymphocytes.
c- Progenitor cells.
d- Mast cells.

11- Extracellular matrix of the connective tissue of the pulp is composed of:

a- Collagen fibers only.
b- Ground substance only.
c- Both 1&2.
d- Non of the above.

12- Pericytes are contractile cells that surround the endothelial cells of:

a- Arterioles.
b- Venules.
c- Lymph vessels.
d- Capillaries.

13- Pulp tissue has receptors for:

a- Heat.
b- Pain.
c- Pressure.
d- All of the above.

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Answers of MCQ

PULP

1	c	8	d
2	b	9	c
3	c	10	a
4	c	11	c
5	b	12	d
6	c	13	b
7	a		

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Periodontal ligament

1- The PDL fibersrunningbetweentwoadjacentteeth:

a-Belongtothe dento-alveolargroup.
b-Are gingival fibers.
c-Are transseptal fibers.
d- Are oxytalan fibers.

2- Which of the following is not true of the alveolodental periodontal fibers? They:

a- Are organized into bundles with differentorientation.
b- Are derived from epithelial rests of Malassez.
c-Undergo continuous remodeling.
d- Are formed and partially resorbed by fibroblasts.

3- Which of the following is not present in the periodontal ligament?

a-Epithelial rests ofMalassez.
b- Fibroblasts.
c- Cementoblasts.
d- Osteocytes.

4- Which of the following principal fibers groups constitutes the main attachment of the tooth?

a- Oblique group.
b- Apical group.
c- Horizontal group.
d- Alveolar group.

5- Which of the following groups of tissues comprise the periodontium?

a- Alveolar bone,periodontal ligament, cementum.
b- Periodontal ligament, cementum, gingiva.
c- Cellular cementum, periodontal ligament, alveolar bone.
d- Cementum, alveolar bone, periodontal ligament, gingiva.

6- The width of the periodontal ligament:

a- It is thiner when teeth are functioning normally.
b- It is narrowest at the level of the midroot area.
c- It becomes wider with aging.
d- It is of uniform width around multi-rooted teeth.

7- The periodontal ligament develops from:

a- Dental organ. b- Dental sac.
c- Dental papilla. d- Dental lamina.

UMHDental

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8- Elasticity of periodontal ligament is due to:

a- Presence of elasticfibers.
b- Presence of oxytalan fibers.
c- Wavy coarse of collagen fibers.
d- Non of the above.

9- The principal fibers of periodontal ligament are:

a- Elastic. b- Oxytalan.
c- Collagen. d- Precollagen.

10- Function of oxytalan fibers:

a- Support nerves. b- Support blood vessels.
c- Help fibroblasts migration. d- All of the above.

11- Open faced nucleus is found in:

a- Resorptive cells. b- Synthetic cells.
c- Defensive cells. d- Progenitor cells.

12- The periodontal ligament is formed of:

a- Cells only. b- Fibers only.
c- Ground substances only. d- All of the above.

13- The protective function of periodontal ligament is achieved by its:

a- Fibers. b- Blood vessels.
c- Nerves. d- All of the above.

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Answers of MCQ

PERIODONTAL LIGAMENT

1	c	8	c
2	b	9	c
3	d	10	d
4	a	11	b
5	d	12	d
6	b	13	d
7	b		

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Bone structure and alveolar process

1- Whichofthefollowingbestdescribesthealveolarboneproper?

a-It is composed of lamellar bone only.
b-It is composed of bundle bone and lamellar bone.
c-It is composed entirely of bundle bone.
d- It is composed of spongy bone only.

2- The alveolar bone proper is composed of:

a- Spongy & woven bone.
b- Bundle & lamellar bone.
c- Compact bone only.
d- Compact & spongy bone.

3- Which of the following is <u>true</u>:

a-Normal long bone contains woven bone.
b-Reversal lines represent rhythmic apposition of bone.
c-Growth line of bone disappears at 5 years.
d- Lamellated bone contains resting lines.

4- Which of the following is <u>wrong</u>?

a- The cribriform plate is another name for alveolar bone proper.
b- Sharpey's fibers are present in bundle bone.
c- Bundle bone develops from the dental sac.
d- Bundle bone is not resorbed.

5-Which of the following <u>best</u> describes the alveolar bone proper?

a- Woven bone and spongy bone.
b- Lamellar bone and bundle bone.
c- Bundle bone and cancellous bone.
d- Woven bone and lamellar bone.

6- Which of the following is <u>not</u> related to the incremental lines:

a- Resting lines.
b- Hunter-schreger bands.
c- Lines of Von Ebner.
d- Retzius lines.

7- Which of the following is not universal component of all hard tissues in man?

a- Collagen.
b- Calcium.
c- Phosphorous.
d- Water.

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8- The major difference between cellular cementum and bone is the:

a- Blood supply.
b- Cellularity.
C- Presence ofSharpey'sfibers.
d- Organic matrix.

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Answers of MCQ

BONE

1	b	5	b
2	b	6	b
3	d	7	a
4	d	8	a

HMHDental

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Oral mucous membrane

1- Sulcularepitheliumisformedof:

a-Nonkeratinized stratified squamous epithelium. b-Parakeratinized stratified squamous epithelium. c-Orthokeratinized stratified squamous epithelium. d-Columnar epithelium.

2- Which of the following is a wrong statement about the dentogingival junction?

a-Ithasan epithelial and a connectivetissue component.
b-Itshifts apically with age.
c-Itscoronal end corresponds to the bottom of the gingival sulcus.
d-Itsepithelial cells are keratinized stratified squamous.

3- Keratinized epithelium differs from nonkeratinized epithelium in that:

a-Itcontains granular cell layer.
b-Containa higher glycogen content.
c-Itdoesnot contain basal lamina.
d-Notonofilaments are found in it.

4- The buccal mucosa:

a-Hasakeratinized epithelium.
b-Is continuous with the gingival mucosa.
c-It isfirmly attached to the underlying tissues.
d-Hasnumerous connective tissue papillae.

5- Masticatory mucosa has:

a-Elasticfibers in its lamina propria.
b-Elasticfibers in its blood vessels wall.
c- Stratum corneum.
d-No connective tissue papillae.

6- Which of the following is true of the sulcular epithelium?

a-Consists ofkeratinizedepithelium.
b-Has submucosa.
c-Is nonkeratinized.
d-Contain connective tissue papillae.

7- Parakeratinized epithelium is found in:

a- Gingiva.
b-Vestibular fornix.
c-Floor of the mouth.
d-Ventral surface of tongue.

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8- In which area the connective tissue papillae are long & numerous?

a-Buccal mucosa.
b- Gingiva.
c-Alveolar mucosa.
d- Floor of the mouth.

9- The mucogingival junction is the junction of:

a-Sulcular epithelium & free gingiva.
b-Free gingiva and attached gingiva.
c-Alveolar mucosa and attached gingiva.
d- Palatine gingival & rougae zone.

10- The mucosa of the dorsal surface of the tongue is:

a-Spesialized mucosa.
b-Masticatory mucosa.
c-Linning mucosa.
d-Smooth surface.

11- Which of the following represents the <u>correct</u> innervation of fungiform papilla?

a-Chorda tympani.
b-Facial nerve.
c-Trigeminal nerve.
d- Glossopharyngeal nerve.

12- Which of the following appears to play an important role in attachment of its cells:

a- Keratohyalinegranules.
b- Desmosomes.
c-Glycogen contents.
d- Lysosomes.

13- The structure of the oral mucosa is:

a-Epithelium andbasement membrane.
b-Mucosa & submucosa.
c-Epithelium, lamina propria and muscular layer.
d-Epithelium, lamina propria and basement membrane.

14- Healthy line is represented in:

a-Mucogingival junction.
b-Free gingival groove.
c- Junction of the palatine gingiva and the lateral surface of the palate.
d- Junction of the buccal and labial mucosa.

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15- Fungiform papillae:

a-Are seenas whitespots.
b-Have a stratum corneum.
c-Carry taste buds.
d- Have a lamina propria which is poorly vascularized.

16- Which of the following is <u>true</u> regarding the dentogingival junction?

a- It is derivedentirely from ectomesenchyme.
b- It moves apically with age.
c- It moves occlusally with age.
d- Contains clear submucosa.

17- Which of the following regions has a non-keratinized epithelium and is loosely attached to the underlying structures?

a- Buccal mucosa.
b-Attached gingiva.
c-Hard palate.
d-Vestibular fornix.

18- The floor of the mouth is lined by:

a-Linning mucosa loosely attached.
b-Linning mucosa firmly attached.
c-Specialized mucosa.
d- Masticatory mucosa.

19- Keratohyaline granules in epithelial cells of keratinized mucosa are associated with:

a- The tonofilaments.
b-Prickle cell layer.
c-Basal cell layer.
d-Stratum granulosum.

20- Which of the following contributes to the reddish coloration of the vermilion border?

a- Thichness of oral epithelium.
b- Existance of high number of langerhans cells.
c- Vascularity of the lamina propria.
d- Epithelial keratinization.

21- Which of the following are <u>not</u> usually found in the keratinocytes of the stratum spinosum of keratinized oral epithelium?

a- Keratohyalinegranules
b- Tonofibrils.
c- Membrane coating granules.
d- Ribosomes.

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22- The sulcular epithelium:

a- Is keratinized.
b- Has thick epithelial layer.
c- Has long epithelial ridges.
d- Is nonkeratinized.

23- Taste buds are <u>not</u> found on:

a- Filliform papillae.
b- Fungiform papillae.
c- Foliate papillae.
d- Circumvallate papillae.

24- Which of the following is <u>not</u> present in the stratum spinosum?

a- Desmosomes.
b- Tonofilaments.
c- Keratohyaline granules.
d- Odland's bodies.

25- Which of the following mucosa has a non-keratinized epithelium :

a- Freegingiva.
b- Attached gingiva.
c- Mucosa of the floor of the mouth.
d- Palatal mucosa.

26- Which of the following is a transiant cell in the oral epithelium:

a- Melanocyte.
b- Lymphocyte.
c- Merkel cell.
d- Langerhans cell.

27- Which of the following serves as a source of the cells which form the attachment epithelium?

a- Dental sac.
b- Reduced enamel epithelium & gingiva.
c- Enamel cuticle.
d- Vestibular lamina.

28- The mode of attachment of the junctional epithelium is:

a- Basement membrane.
b- Basal lamina & hemidesmosome.
c- Enamel cuticle.
d- Odland's bodies.

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29- Which of the following is true of the attachment epithelium?

a- Its coronal end is found at thebottomofthegingivalsulcus.
b- Its apical migration is the major causative factor in active eruption.
c- Is impermeable to dyes and fluids.
d- Can migrate coronally.

30- Which of the following cells synthesize protein which remains within the cell without secretion:

a- Osteoblasts.
b- Plasma cell.
c- Fibroblast.
d- Keratinocyte.

31- Which of the following papillae of the tongue do not contain taste buds:

a- Filliform.
b- Vallate.
c- Foliate.
d- Fungiform.

32- Which is not true about taste buds:

a- The supporting cells of a taste bud carry taste hairs.
b- 10 to 20 neuroepithelial cells are the receptors of taste stimuli in a taste bud.
c- Taste buds extend from the basal lamina to the surface of epithelium.
d- Nerve fibers end in contact with the sensory cells in the taste buds.

33-Those papillae of the tongue which are the largest, having many taste buds and are associated with the duct of Von Ebner's gland are:

a- Folliate.
b- Filliform.
c- Fungiform.
d- Circumvallate.

34- Glands emptying the circumvalate trough are the:

a- Mucous glands.
b- Serous glands.
c- Glands of Nuhn.
d- Mucous and serous mixed glands.

35- Tongue papillae responsible for bitter taste sensation:

a- Filliform papillae.
b- Fungiform papillae.
c- Circumvallate papillae.
d- Foliate papillae.

HMU Dental

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Answers of MCQ

ORAL MUCOSA

1	a	19	d
2	d	20	c
3	a	21	a
4	c	22	d
5	b	23	a
6	c	24	c
7	a	25	c
8	b	26	b
9	c	27	b
10	a	28	b
11	a	29	a
12	b	30	d
13	d	31	c
14	a	32	a
15	c	33	d
16	b	34	b
17	d	35	c
18	a		

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Salivary gland

1- Mucous cells of salivaryglandshave:

a-Rounded nuclei.
b-Flattened nuclei.
c-Form the cap of the demilunes.
d-Have zymogen granules.

2- Which of the following is <u>not</u> function of the striated duct?

a-Iodine secretion.
b-Zymogen secretion.
c-Sodium & chloride transport.
d- Potassium & calcium transport.

3-Which of the following is <u>not</u> characteristic of serous cells of a salivary gland?

a-The cells are pyramidal in shape.
b-Its secretory granules are called zymogen granules.
c-They are capped by mucous cells in a mixed salivary acinus.
d-There is junctional complexes between the secretory cells.

4- Which of the following adult gland is <u>incorrectly</u> paired with its secretion?

a-Parotid - pure serous.
b-Palatal salivary glands - pure mucous.
c-Submandibular - mixed salivary gland.
d- Sublingual - pure serous.

5- The secretory canaliculi in salivary gland is found in:

a-mucous acini.
b-Striated ducts.
c-Serous acini.
d-Main duct.

6- The excretory ducts of the major salivary glands are lined with:

a-Simple cubiodal epithelium with gobletcells.
b-Simple columnar without goblet cells.
c-Stratified squamous epithelium with goblet cells.
d- Pseudostratified columnar epithelium with goblet cells.

7- The cytoplasmic basophilia observed in acinar cells of the adult parotid gland is due to:

a- Mucigen granules.
b- Ribosomes.
c- Glycoproteins.
d-Lysosomes.

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8- Pure mucous salivary gland is:

a-Parotid gland.
b-Major sublingual glands.
c-Glossipalatine gland
d-Submandibular gland.

9- The basophilia observed in acinar cells of the parotid gland is due to:

a- Mucigen granules.
b- Lysosomes.
c- Glycoproteins.
d- Ribosomes.

10- In the demilune of the mixed salivary acinus, the mucous secretory cells:

a- Have rounded nuclei.
b- Form demilune cappings.
c- Have zymogen granules.
d- Line the luminal surface of the acinus.

11- The concentration of sodium and chloride in saliva is controlled by:

a- Highly differentiatedserous cells.
b- Well developed mucous cells.
c- Striated ducts.
d- Intercalated ducts.

12- The role of the myoepithelial cells in the salivary glands is:

a- Defensive mechanism.
b- Contraction of the acinus.
c- Prevention of collapse of the acini.
d- Prevention of extravasation of saliva into the glandular stroma

13- Which of following is <u>most</u> likely to be a product of a salivary gland serous cell?

a- Collagenase enzymes.
b- Mucous secretion.
c- Synovial fluid.
d- Enzyme which digests starch and other polysaccharides_

14- Which of the following glands has a purely serous secretion?

a- Labial.
b- Anterior lingual.
c- Submandibular.
d- Von Ebner's.

LWHDental

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15- The secretions of salivary glands are:

a- Exocrine and holocrine.
b- Exocrine and merocrine.
c- Exocrine and apocrine.
d- Endocrine and merocrine.

16- Which minor salivary glands are purely serous:

a- Lingual glands of VonEbner.
b- Lingual glands of Blandin Nuhn.
c- Palatine glands.
d- Labial and buccal glands.

17- A serous acinar cell has all the following characteristic except:

a- Numerous secretory granulesin apicalcytoplasm.
b- Cytoplasm is abundant and basophilic.
c- Basal cell membrane has many infolding between which rows of mitochondria are lodged.
d- Extensive rough endoplasmic reticulum is present.

18- The groups of entirely serous adult salivary glands are:

a- VonEbner's and palatal.
b- Parotid and Von Ebner's.
c- Parotid and submandibular.
d- Sublingual and submandibular.

19-The ducts of the salivary glands that are lined by a single layer of tall columnar cells with radially arranged mitochondria are known as:

a- Lobarducts.
b- Striated ducts.
c- Intercalated ducts.
d- Interlobar ducts.

20- Large cells with a small pyknotic nucleus and abundant cytoplasm, mainly inthe parotid and submaxillary glands are the:

a- Myoepithelialcells.
b- Oncocytes.
c- Serous cells.
d- Cells of the intercalated ducts.

21- Thin branching tubes of varying lengths that connect the terminal portions with the striated ducts of salivary glands are:

a- Interlobular ducts.
b- Intercalated ducts.
c- Intralobular ducts.
d- lobar ducts.

HMIDental

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22- Which of the following is <u>not</u> true of parotid glands:

a- Stenson'sduct opens in thecheek oppositemaxillarysecond molar.
b- Predominantly serous gland.
c- Long, branching intercalated ducts are present.
d- Serous demilunes cap mucous secretory units.

23- Which of the following is <u>false</u> about sublingual glands:

a- The excretory duct opensat the sublingualcaruncle.
b- It is a predominantly mucous gland.
c- They have poorly developed striated ducts.
d- They have a long branching intercalated duct structure.

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Answers of MCQ

SALIVARY GLANDS

1	b	13	d
2	b	14	d
3	c	15	b
4	d	16	a
5	c	17	c
6	d	18	b
7	b	19	b
8	c	20	b
9	d	21	b
10	d	22	d
11	c	23	d
12	b		

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Maxillary sinus

1- Which of the followingistrueconcerningthemaxillarysinus?

a-It begins to develop atage6 months.
b-Its growth is by pneumotization.
c-It is lined by pseudostratified squamous epithelium.
d-Its opening is in the oral cavity.

2- Maxillary sinus open in the:

a- Medial nasal conche.
b- Superior nasal meatus.
c- Inferior nasal meatus.
d- Medial nasal meatus.

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Answers of MCQ

MAXILLARY SINUS

1	b	2	d

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Tempro-mandibular joint

1- The TMJ does<u>not</u>contain:

a-Articular disc.
b-Capsular ligament.
c-Synovial membrane.
d-Blood vessels in its covering fibrous connective tissue.

2- Which is <u>true</u> of the T.M.J?

a-An epithelium lines the synovial membrane.
b-It has a highly vascular articulator disc.
c-The articular surfaces are covered by a dense fibrous C.T.
d-The articular disc is attached to the roof of the glenoid fossa.

3- The head of the mandibular condyle:

a-Articulates with the roof of the glenoid fossa.
b-Develops from the secondary cartilage.
c-Lacks hyaline cartilage.
d- Consists of spongy bone only.

4-Which of the following statement about TMJ is <u>true</u>:

a- TMJ is a bilateral synovial articulation between the mandible and the cranium.
b- TMJ is a unilateral synovial articulation between the mandible and the cranium.
c- TMJ is a unilateral synovial articulation between the maxilla and the cranium.
d- TMJ is a bilateral synovial articulation between the maxilla and the cranium.

5- TMJ development starts at:

a- 3 MIU.
b- 4 MIU.
c- 5 MIU.
d- 6 MIU.

6- TMJ Condyle is:

a-Strongly convex in the antroposterior direction & slightly convex mediolaterally.
b- Strongly convex in the mediolateral direction & slightly convex antroposteriorly.
c- Concavo-convex.
d-Concave in the antroposterior direction & convex in the mediolateral direction.

7-The secondary cartilage present in the head of the TMJ condyle remains active till:

a- The end of the 3rd decade.
b- The beginning of the fourth decade.
c- The end of the second decade.
d- The beginning of the second decade.

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8- Glenoid fossa is:

a- A depression on the inferior surface of the squamous part of the temporal bone.
b- An elevation on the inferior surface of the squamous part of the temporal bone.
c- A depression on the anterior surface of squamous part of the temporal bone. d- An elevation on the inferior surface of the zygomatic part of the temporal bone.

9- The glenoid fossa is limited:

a- Medially by the spine of sphenoid & laterally by the root of the zygomatic process of the temporal bone.
b- Laterally by the spine of sphenoid & Medially by the root of the zygomatic process of the temporal bone.
c- Medially by the root of the zygomatic process of the temporal bone & laterally by the spine of sphenoid.
d- Medially by the spine of sphenoid & laterally by the squamous part of the temporal bone.

10- TMJ articular eminence is:

a- Present at birth.
b- Absent at birth.
c- Formed during birth.
d- Present at childhood.

11- TMJ disk is:

a- Biconcave of uneven thickness.
b- Concavo-convex of uneven thickness.
c- Biconcave of even thickness.
d- Concavo-convex of even thickness.

12- One of the following is not the exact function of TMJ ligament:

a- Prevent lateraldisplacement of one joint&medialdisplacementof the other.
b- Limit the amount of inferior displacement.
c- Prevents posterior displacement.
d- Restricts the TMJ displacement in all planes.

13- The TMJ large marrow spaces:

a- Remain unchanged.
b- Increases by age.
c- Decrease by age.
d- Richin collagentypel

14- TMJ collagen fibers show alternating dark & light bands when viewed in polarized light which indicate that:

a- They are wavyorcrimped.
b- They are straight.
c- They are twisted.
d- They are corrugated.

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15- Condylar cartilage cells:

a- Increase in number with age.
b- Decrease in number with age.
c- Remain unchanged.
d- Degenerate with age.

16- The hyaline cartilage in the condyle serves as an active growth center till the age of:

a- 30 years.
b- 10 year.
c- 20 years.
d- 40 years.

17- The periodicity of the crimping seen in TMJ collagen fibers is of the order of:

a- 30-40 μm.
b- 15-20 μm.
c- 10-25 mm.
d- 15-30 mm.

18- The ground substance of TMJ disk comprises about:

a- 10% of its dry weight.
b- 5% of its dry weight.
c- 12% of its dry weight.
d- 8% of its dry weight.

19- Which of the following statements about TMJ intra- articular disk is true:

a- About ⅔ of glygosaminoglycans is chondroitin sulphate & about ⅓ is dermatan sulfate.
b- About ⅓ of glygosaminoglycans is chondroitin sulphate & about ⅔ is dermatan sulfate.
c- About ⅛ of glygosaminoglycans is chondroitin sulphate & about ⅜ is dermatan sulfate.
d- About ⅜ of glygosaminoglycans is chondroitin sulphate & about ⅛ is dermatan sulfate.

20-The TMJ disk is:

a- Avascular & not innervated in its central region.
b- Vascular & innervated.
c- Avascular & innervated.
d- Vascular & not innervated.

21- At rest the hydrostatic pressure of TMJ synovial fluid has been reported as being:

a- Atmospheric.
b-Supra-atmospheric.
c- Sub-atmospheric.
d- Sometimes sub-atmospheric & other times atmospheric.

22- Which one of the following is <u>not</u> a function of the TMJ synovial fluid:

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a- Lubricatiuon. c- Debridement. b- Nourishment.
d- Pressure dampening.

a- External carotid artery. b- Internal carotid
artery. c- Common carotid artery. d- Both
External & Internal carotid arteries.

23- The vascular supply to the TMJ comes from:

24- The TMJ takes branches from:

a- The mandibular division of the trigeminal nerve.
b- The maxillary division of the trigeminal nerve.
c- The ophthalmic division of the trigeminal nerve.
d- The facial branch of the mandibular nerve.

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Answers of MCQ

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TEMPRO-MANDIBULAR JOINT

1	d	13	c
2	c	14	a
3	a	15	a
4	a	16	c
5	a	17	b
6	a	18	b
7	c	19	a
8	a	20	a
9	a	21	c
10	b	22	d
11	a	23	a
12	d	24	a

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